



Dynamic behavior of molten pool and evolution of joint properties in laser full/partial penetration welding of copper under alternating magnetic field modulation



Zhiheng Zeng ^a, Lifang Mei ^{a b c} , Dongbing Yan ^{a c} , Qingying Dai ^a, Yang Liu ^a

Show more

Outline | Share Cite

<https://doi.org/10.1016/j.optcom.2025.131853>

[Get rights and content](#)

Full text access

Highlights

- Effect of alternating magnetic field on laser full- and partial-penetration welding of copper was studied.
- The temperature, flow velocity and keyhole shape of the full-penetration welding pool are better.

- The grain size of the full-penetration welding is more refined than that of the partial-penetration.
- The comprehensive performance of the full-penetration weld is higher than that of the partial-penetration.

Abstract

To address the low energy coupling efficiency and poor process stability in T2 red copper laser welding, the effect of the alternating magnetic field on the weld pool properties and joint quality of T2 red copper laser welding was investigated by numerical simulation and experiment. The results show that the application of alternating magnetic field can effectively improve the temperature of the weld pool, the fluid flow rate and the stability of the keyhole, promote the fine grain structure inside the weld, and thus improve the mechanical and conductive properties of the joint. After applying the alternating magnetic field, the temperature difference of the molten pool of the full-penetration weld was significantly reduced (847K), which was much lower than that of the partial-penetration weld (2840K), while the maximum flow velocity of the molten pool (2.217m/s) was higher than that of the partial-penetration weld (1.932m/s). Additionally, in the full-penetration state, the keyhole depth exhibits smaller fluctuations, the oscillation duration is shorter, and the simulated weld fusion line morphology shows high agreement with the experimental results. Regarding joint properties, the microhardness, tensile strength, and electrical conductivity of the full-penetration weld reached 68.76 HV, 205MPa, and 65.31 MS/m, respectively, representing increases of 7.71%, 9.63%, and 8.54% compared to the partial-penetration weld. Under the alternating magnetic field, the effective heat input from the laser to the base material is significantly enhanced, thus regulating the hydrodynamic behavior of the molten pool more effectively, especially in the full-penetration welding process. This not only improves the quality of weld formation, but also enhances the overall welding performance, which provides a new basis and revelation for the theoretical research and technical application of T2 red copper laser welding.

Keywords

Alternating magnetic field-assisted laser welding; T2 red copper; Numerical simulation; Molten pool; Full-penetration; Partial-penetration

1. Introduction

Due to its superior electrical conductivity, thermal conductivity, and ductility, the demand for T2 red copper components is increasing in aerospace, new energy vehicles, lithium copper foil, and electronics. With advancements in welding technology and quality control for T2 red copper weldments, traditional welding methods such as friction stir welding, brazing, and arc welding often result in a coarse microstructure in the fusion zone, leading to a reduction in the mechanical properties and electrical conductivity of the welded joints. Consequently, these methods struggle to meet the high-precision and high-performance connection requirements of modern automotive manufacturing [1,2]. In contrast, laser welding has shown greater potential for application in red copper components because of its advantages [3,4], including excellent flexibility, a large weld depth-to-width ratio, a narrow heat-affected zone, minimal post-weld deformation, high precision, and high efficiency [5]. Due to the high reflectivity and thermal conductivity of red copper, laser energy absorption is reduced, resulting in an unstable welding process prone to defects such as spattering, porosity, hump formation, and biting edges, which negatively impact weld quality [6]. Therefore, improving energy coupling efficiency and process stability in red copper laser welding is a key challenge. To address this, extensive studies have been conducted on enhancing weld quality.

In recent years, many scholars have focused on improving the laser absorption efficiency of copper through laser modulation, surface pretreatment, and the use of dual-beam lasers. For example, Liang et al. [7] demonstrated that circular laser oscillation welding significantly improves the quality of T2 red copper welds within a specific range of oscillation process parameters. Furthermore, the oscillation of the laser beam helps to increase the melt width of the bonding surface, thereby enhancing the mechanical properties of the welded joints. Andreas et al. [8] found that sinusoidal power modulation can significantly reduce the amount of melt ejection, which improves keyhole stability, thereby enhancing weld depth and surface morphology. However, since sinusoidal power modulation primarily controls the laser output power characteristics, its mode of action is more complex, which results in certain limitations for practical applications. Jiao et al. [9] used a nanosecond laser with a wavelength of 532nm to pre-treat the surface of red copper, successfully reducing its reflectivity to near-infrared laser light from 95% to 15.5%,

significantly improved the efficiency of laser energy utilization in the welding process. Furthermore, studies have demonstrated that surface pretreatment can significantly enhance the overall performance of materials [[10], [11], [12]]. Chen et al. [13] observed that surface treatments such as sandblasting, blackening, and nanocomposite addition enhanced welding efficiency in laser welding of high-reflectivity pure copper, with nanocomposites having the most significant effect. However, the formation of intermetallic compounds may adversely affect the mechanical properties of the weld. Wang et al. [14] pointed out that the use of a dual-wavelength laser beam to weld pure copper sheets can significantly improve the surface morphology and forming quality of the weld. However, this method complicates the control of energy distribution from the heat sources during welding.

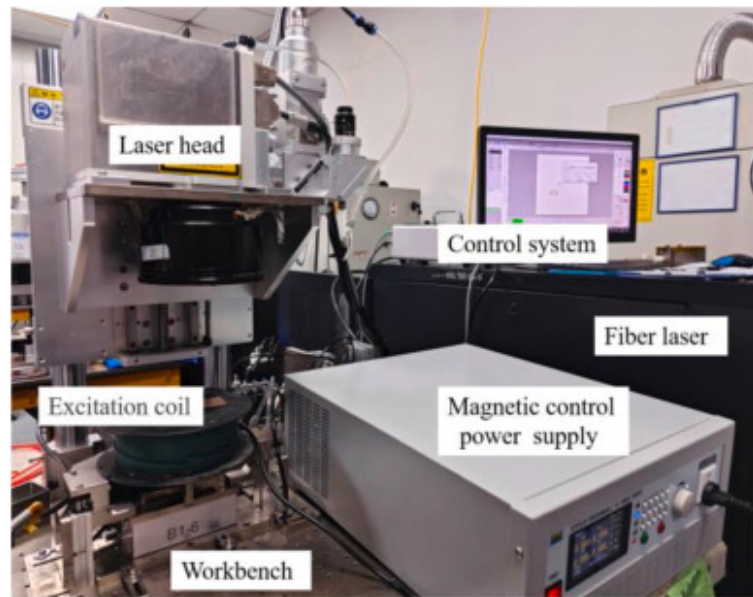
Magnetic field-assisted laser welding, which improves weld quality through the electromagnetic effect, has received widespread attention and rapid development. This technology not only reduces laser energy loss in the molten pool and improves heat input, but also effectively inhibits porosity defects and enhances weld quality [15,16], providing a new approach for high-quality laser welding of red copper. Therefore, scholars have conducted numerous studies on magnetic field-assisted laser welding. For example, Liang et al. [17] integrated numerical simulation with experiments to reveal the Lorentz force generated by the interaction between the steady-state magnetic field and the molten pool and its effect on molten pool flow. Cao et al. [18] developed a three-dimensional numerical model to reveal the mechanism of the magnetic field's influence on the weld pool. The results showed that, under the influence of the magnetic field, the molten pool generates a Lorentz force opposite to the fluid flow, inhibiting fluid flow toward the edge, reducing heat transfer from the center to the edge of the molten pool, and lowering the temperature gradient. Chen et al. [19,20] developed a three-dimensional numerical model incorporating various physical mechanisms, including magnetohydrodynamics and electromagnetic effects, to analyze the molten pool behavior in magnetic field-assisted laser welding. The results indicate that the Lorentz force generated by the electromagnetic field facilitates uniform temperature distribution in the molten pool, reduces laser energy loss during welding, and improves weld quality. Hu et al. [21] used alternating magnetic field-assisted laser welding of red copper-steel dissimilar materials to study the effect of the alternating magnetic field on the microstructure, elemental distribution, and mechanical properties of the welded joints. Liu et al. [22] studied the effect of the applied alternating magnetic field on the microstructure and mechanical properties of 361 stainless steel laser-MIG hybrid welded joints. The results show that the stirring effect of the alternating magnetic field can effectively improve the solidification time of the molten pool, promote austenite grain refinement, and thus enhance the mechanical properties of the joint. Meng et al. [23] demonstrated that after applying an appropriate magnetic field strength, the grain structure in the upper part of the weld was significantly refined. The Lorentz force generated in the molten pool due to electromagnetic stirring positively affects the flow of molten metal and solidification. Zhu et al. [24] showed that the alternating

magnetic field altered the morphology of the droplets via the Lorentz force and effectively suppressed spattering, thereby optimizing droplet deflection and improving welding quality and efficiency. Liu et al. [25] investigated the effect of different magnetic field waveforms on aluminium alloy welding. The results showed that all three waveforms improved the weld depth-to-width ratio and reduced porosity, attributed to enhanced keyhole stability, prolonged solidification time, and better gas bubble escape. Bachmann et al. [26] found that applying an alternating magnetic field with a certain strength during laser welding accelerates gas bubble escape in the molten pool, significantly reducing porosity. This effect is primarily due to the combined action of the electromagnetic Archimedean force and molten pool stirring.

Previous studies indicate that applying an alternating magnetic field with a specific strength during welding significantly improves weld quality. This is mainly due to electromagnetic stirring, which refines the grain structure, enhances mechanical properties, and reduces defects like porosity and cracking. However, current research on alternating magnetic field-assisted laser welding mainly focuses on metals like magnesium, aluminium, and steel, with limited studies on T2 red copper laser welding. To enhance the energy coupling efficiency and poor process stability in T2 red copper laser welding, this paper combines numerical simulation and experimental studies to investigate the impact of an alternating magnetic field on the melt pool behavior and weld quality in full-penetration (1 mm) and partial-penetration (2 mm) specimens. Firstly, an experimental study on T2 red copper laser welding assisted by an alternating magnetic field was conducted to analyze the effect of alternating magnetic field with different frequencies and intensities on the morphology of weld seam in both full- and partial-penetration states, with the aim of determining the optimal magnetic field parameters. Subsequently, a three-dimensional thermal-fluid coupling numerical model of T2 red copper laser welding was established based on the experimental parameters to compare the effects of alternating magnetic field strength on the temperature distribution, fluid flow, and keyhole morphology of the specimens in both full- and partial-penetration states. The melt pool morphology predicted by the model was validated by comparison with experimental results. Finally, the effects of the alternating magnetic field on the microstructure, mechanical properties, and electrical conductivity of the welded specimens in both full- and partial-penetration welds are analyzed. Through the above analysis, the mechanism by which the alternating magnetic field affects full- and partial-penetration welding is elucidated, providing both an experimental basis and theoretical support for optimizing the laser welding process of T2 red copper and its alloys.

2. Research conditions and methods

The main equipment used for the welding experiments is the IPG YLR-2000-SM single-mode fiber laser, with a rated power output of 2000 W. The laser welding head is fitted with a CL-H201C-140–330 diaphragm, a laser spot diameter of 0.15 mm, a focusing lens of 200mm focal length, and a laser wavelength of 1064nm. The test platform is shown in Fig. 1. In this paper, the welding test materials consist of T2 red copper plates with thicknesses of 1 mm and 2 mm, and their chemical composition and mechanical properties are listed in Table 1.



[Download: Download high-res image \(460KB\)](#)

[Download: Download full-size image](#)

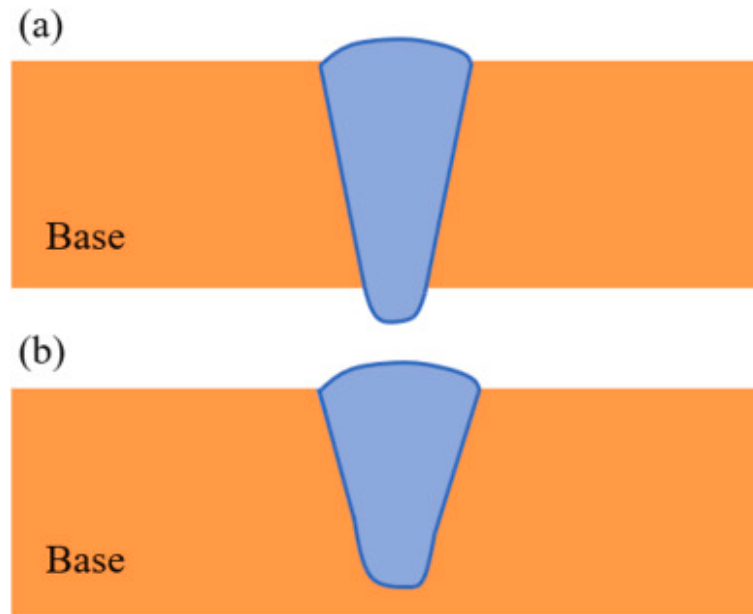
Fig. 1. Laser welding experimental platform.

Table 1. Chemical composition (mass fraction, %) and mechanical properties of T2 red copper [27,28].

Element	Cu	Fe	Pb	S	Sb	Bi	As
Mass fraction/%	≥99.9	≤0.005	≤0.005	≤0.005	≤0.002	≤0.001	≤0.001
Mechanical	Tensile strength/MPa		Harness/HV		Elongation/%		

Element	Cu	Fe	Pb	S	Sb	Bi	As
Properties	295		90		3		

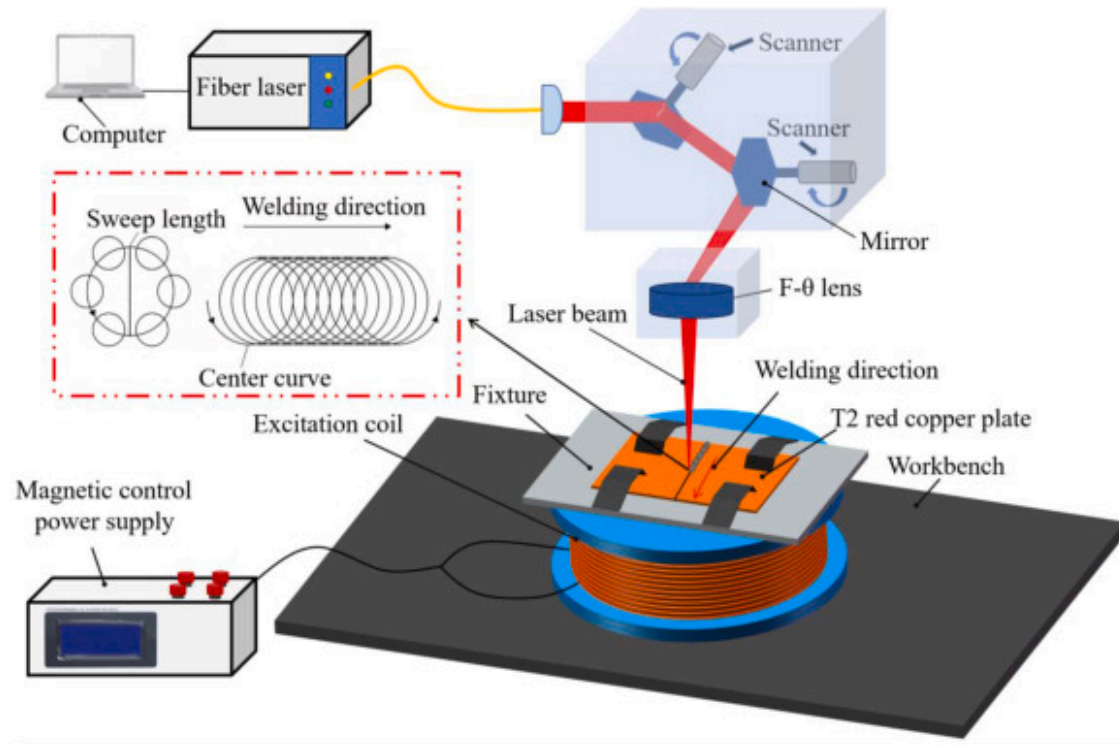
After a series of preliminary experiments, a set of optimal laser welding process parameters without the magnetic field was selected: laser power $P=1700\text{W}$, welding speed $v=35\text{mm/s}$, defocus amount $\Delta f=0\text{mm}$, and oscillation frequency $f=250\text{Hz}$. Keeping this set of parameters constant, experiments on the laser welding of T2 red copper are conducted under different magnetic field frequencies and strengths. Furthermore, the states of the full- and partial-penetration are shown in [Fig. 2](#). Before welding, firstly, use sandpaper to polish the area to be welded with T2 red copper plate to remove burrs and ensure that the docking surface is smooth and even; then wipe it with anhydrous ethanol to remove the surface impurities; subsequently, use the fixture to clamp and fix the two pieces of T2 red copper butt joints. Finally, the excitation coil was positioned vertically under the fixture. The frequency and strength of the alternating magnetic field in the excitation coil were changed by adjusting the excitation frequency and current. A high-precision Tesla meter was used to measure the magnetic field's frequency and strength at the welding surface of the specimen. The experimental scheme and selected parameters of the alternating magnetic field are shown in [Fig. 3](#) and [Table 2](#), respectively. According to the test results of the alternating magnetic field parameters for process optimization, the weldments in the full- and partial-penetration states the formation of a better quality of a set of alternating magnetic field parameters. Based on the optimized alternating magnetic field parameters, a three-dimensional heat-fluid coupling numerical model for laser welding of T2 red copper was developed. The model incorporated heat source parameters, boundary conditions, and material properties to analyze the effects of the alternating magnetic field on the melt pool temperature distribution, fluid flow, and keyhole morphology in both full- and partial-penetration states. Additionally, the microstructural, mechanical, and electrical conductivity properties of fully welded and partially welded specimens under the influence of the alternating magnetic field were compared and analyzed.



[Download: Download high-res image \(165KB\)](#)

[Download: Download full-size image](#)

Fig. 2. Welding status: (a) full-penetration, (b) partial-penetration.



[Download: Download high-res image \(527KB\)](#)

[Download: Download full-size image](#)

Fig. 3. Schematic diagram of the alternating magnetic field-assisted laser welding of T2 red copper. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 2. Alternating magnetic field parameters.

Alternating frequency (HZ)	Alternating magnetic field strength (mT)
100、150	0
	20
	40

Alternating frequency (HZ)

Alternating magnetic field strength (mT)

60

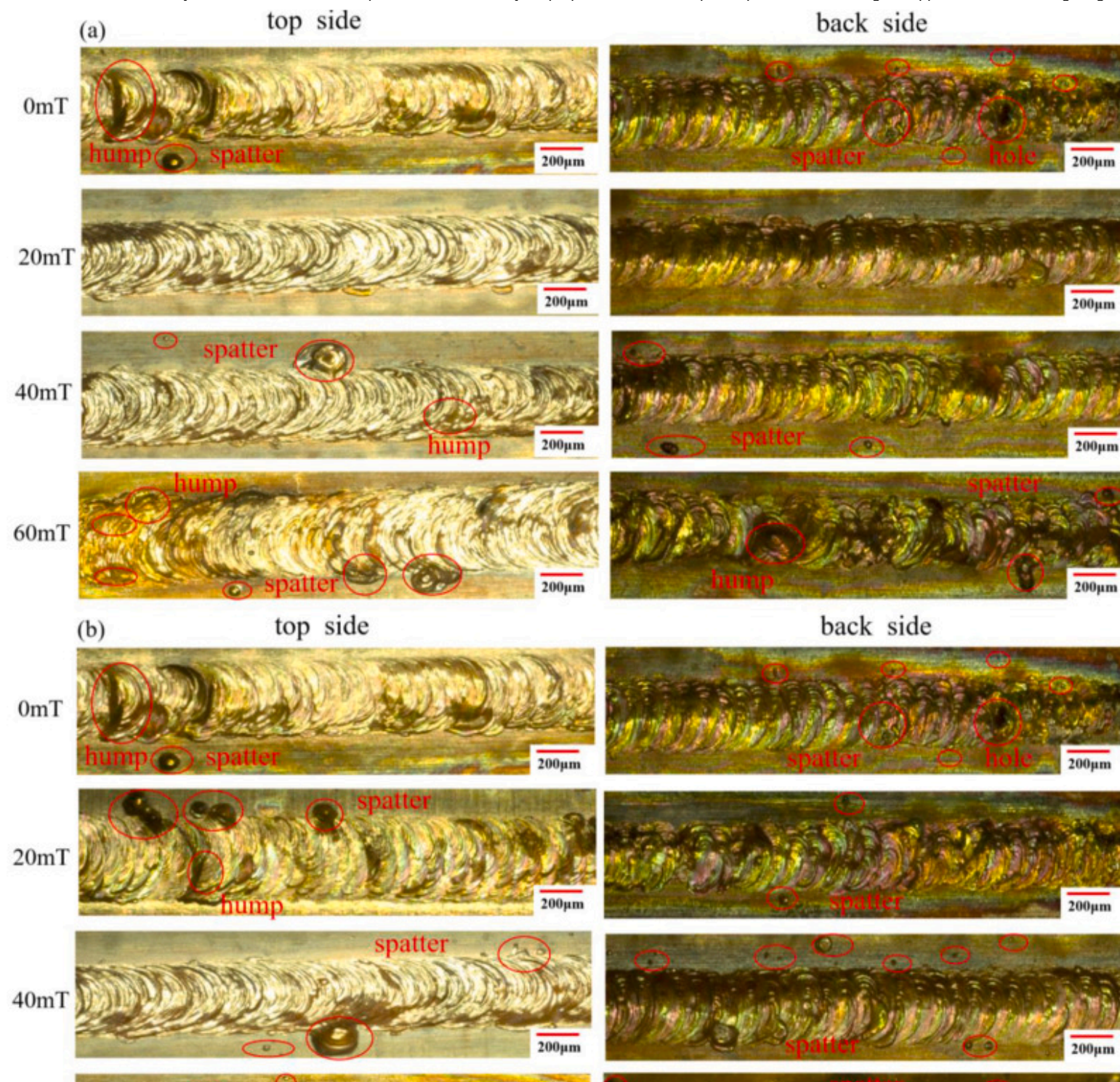
After the completion of the welding experiments, firstly, the VHX 2000 three-dimensional microscope was used to observe the surface morphology of the weld and analyze the appearance of the weld seam morphology and surface defects; subsequently, the welded specimens were prepared into metallurgical specimens by wire-cutting samples, mounting, grinding, polishing and corrosion, and the cross-sectional morphology and metallurgical organization of the specimens were observed under the ZEISS NEOPHOT30 metallurgical microscope. Furthermore, in order to obtain the mechanical properties of the specimen, the HVS-1000A Vickers microhardness tester was used to measure the hardness of the weld in the T2 red copper. The test load was set to 0.1 kgf with a loading time of 15 s. Hardness measurements were taken at five points from the top to the bottom of the weld seam area of each specimen, and the average of these five points was considered as the final microhardness value for the weld seam area. According to the national standard GB/T 34505-2017, "Copper and Copper Alloy Materials Room Temperature Tensile Test Method", the welding pieces are prepared into tensile samples. With WDW-20C microcomputer-controlled electronic universal testing machine, set the tensile speed of 2 mm/min. Tensile tests were carried out on three groups of welded specimens at the same alternating magnetic field strength, and the average of the three tests was taken as the final tensile strength of the specimens at that alternating magnetic field strength. Finally, the conductivity of the full- and partial-penetration joints under different alternating magnetic field strengths was measured using an RTS-9 double electrodiagnostic four-point probe tester. Each specimen was tested three times, and the average of the three measurements was taken as the final conductivity of the welded joints under this alternating magnetic field.

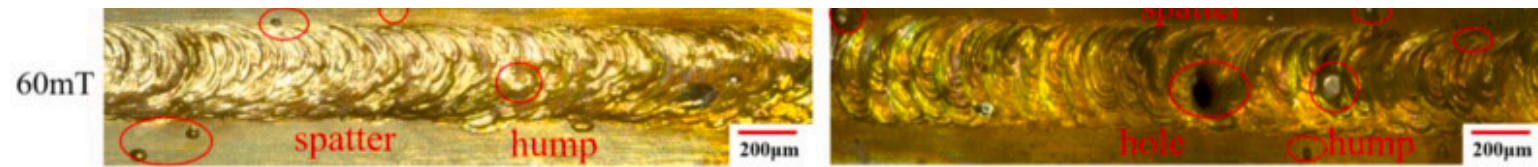
3. Analysis of alternating magnetic field process parameters

3.1. Effect of alternating magnetic field on the morphology of full-penetration welds

Firstly, the effects of varying alternating magnetic field frequencies and strengths on the weld appearance and cross-sectional morphology of the T2 red copper specimens in the fused state were analyzed. The results were compared with those of specimens without an applied magnetic field, as illustrated in [Fig. 4](#), [Fig. 5](#). As illustrated in [Fig. 4](#), in the absence of a magnetic field, the weld seam surface exhibits significant unevenness, with localized collapse and hump formation, there is a high

presence of spatter and a few holes at the back of the weld, resulting in poor overall weld quality. At the same time, as depicted in Fig. 5, without the magnetic field, the liquid metal flows toward the bottom of the molten pool under gravity during welding, leading to a concave upper end and convex lower end in the weld cross-section. Furthermore, as illustrated in Fig. 4, Fig. 5, when the magnetic field frequency is 100Hz, the weld morphology improves initially with increasing magnetic field strength, then deteriorates. At a magnetic field strength of 20mT, the weld seam surface exhibits a uniform, dense, and continuous fish-scale ripple pattern, with no significant spatter, porosity, or other defects. The back of the weld is flat, continuous, and uniform, while the weld cross-section shows flatter upper and lower ends, free from undercutting defects, resulting in improved weld quality. When the magnetic field strength reaches 40mT, the surface forming quality of the weld is improved compared with that of the non-magnetic field, the front section has a fine fish-scale pattern, but there are large particles of metal spatter and hump defects at the edge of the weld, and the back of the weld is also spattered, and the weld cross-section is slightly concave at the upper end and protrudes out at the lower end, and the overall forming quality decreases compared to that of the weld at 20mT. When the magnetic field strength is further increased to 60mT, the weld seam surface becomes uneven, with disordered and irregular corrugations. Hump and spatter defects increase, and the back of the weld exhibits a disorganized and discontinuous appearance. The weld cross-section shows concavity at the upper end and convexity at the lower end, and the overall weld quality is poorer compared to that without the applied magnetic field.

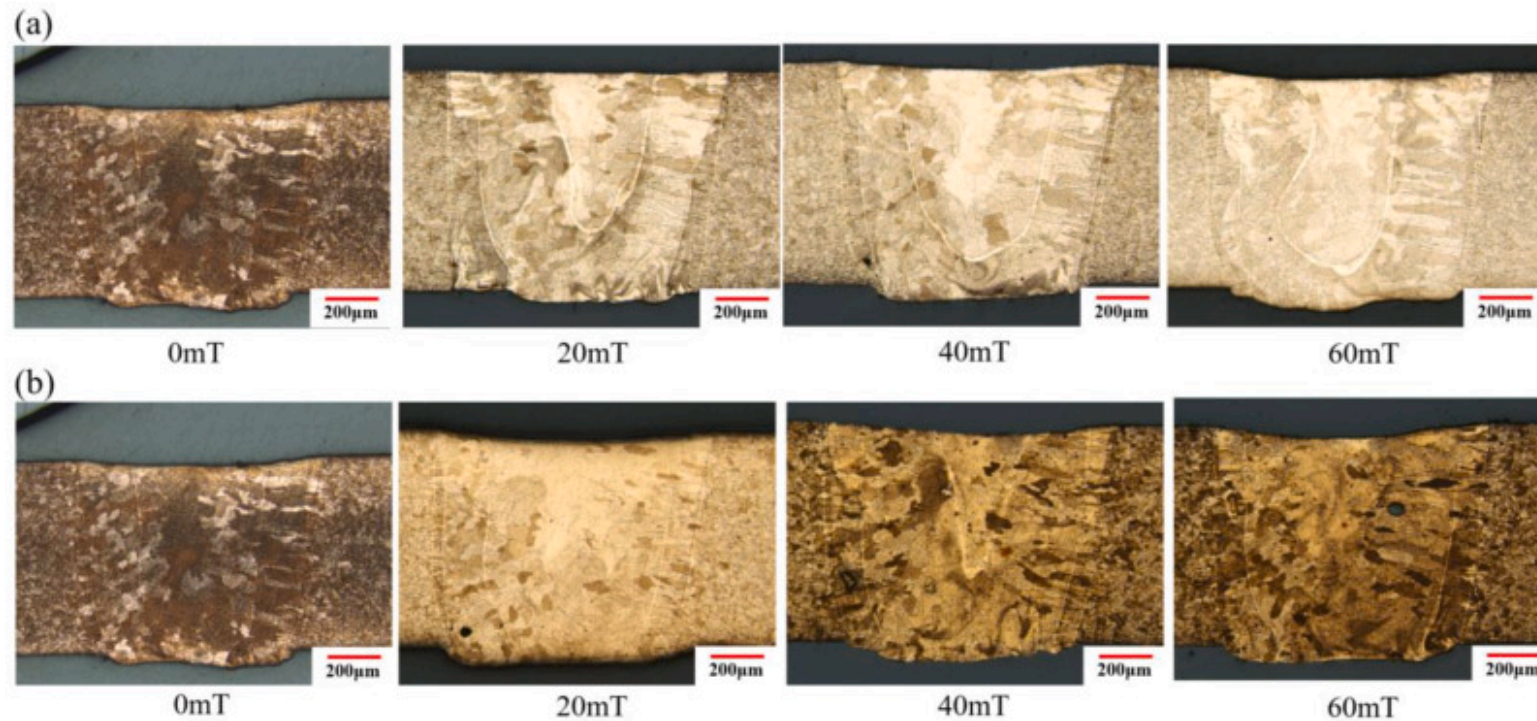




[Download: Download high-res image \(3MB\)](#)

[Download: Download full-size image](#)

Fig. 4. Weld surface morphologies of full-penetration specimens under different alternating magnetic field frequencies and strengths: (a)100Hz, (b)150Hz.



[Download: Download high-res image \(1MB\)](#)

[Download: Download full-size image](#)

Fig. 5. Weld cross-section morphologies of full-penetration specimens under different alternating magnetic field frequencies and strengths: (a)100Hz, (b)150Hz.

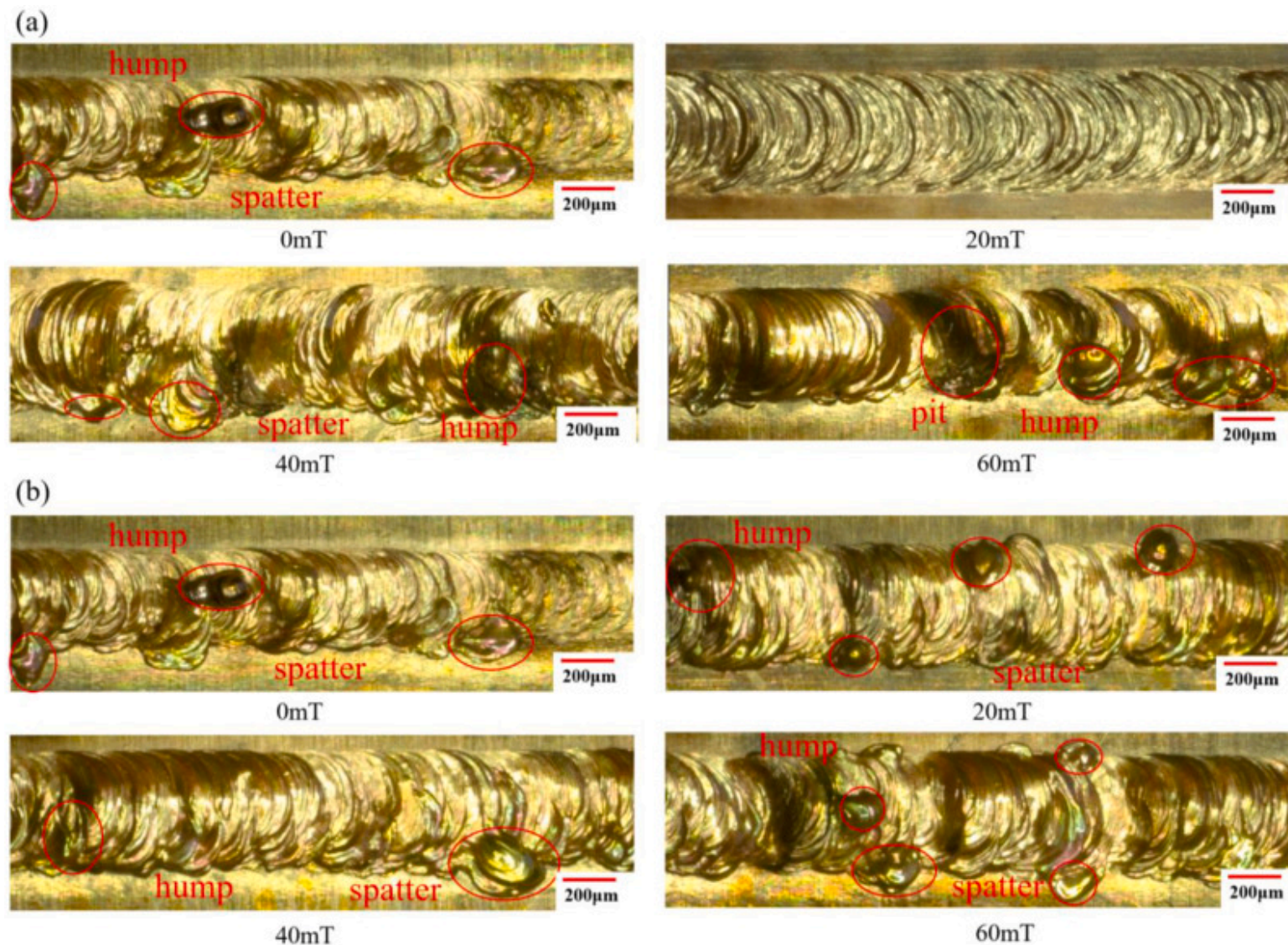
As illustrated in Fig. 4, Fig. 5, when the magnetic field frequency is increased to 150Hz, the trend in weld formation quality follows a pattern similar to that observed at 100Hz. With increasing magnetic field strength, the weld quality improves initially, then deteriorates. At magnetic field strengths of 20mT and 60mT, the weld seam surface and back of the weld exhibit disorganization, with numerous metal spatter particles, humps, and other defects. The weld cross-section shows significant invagination, resulting in poor overall weld formation quality. At a magnetic field strength of 40mT, although individual large metal spatter particles and a small number of smaller spatter particles are present on both the weld surface and the back of the fusion line, the fish-scale ripples on the weld surface and back are more uniform and continuous. The overall weld surface and

cross-sectional morphology are more integrated, and the weld quality is better than that observed in the absence of a magnetic field and at other magnetic field frequencies and strengths.

The above analysis shows that the frequency and strength of the alternating magnetic field have a certain effect on the quality of weld formation under the state of T2 red copper specimen full-penetration. When the magnetic field frequency is set at 100Hz, the magnetic field strength for better weld forming quality is 20mT, and when the magnetic field frequency is set at 150Hz, the magnetic field strength for better weld forming quality is 40mT. As illustrated in [Fig. 4](#), [Fig. 5](#), at a magnetic field frequency of 100Hz and a strength of 20mT, the weld seam profile exhibits a uniform and continuous flat fish-scale ripple pattern, with minimal defects such as spatter and humps; the backside of the weld is also flat and continuous, and the upper and lower ends of the cross-sectional profile are relatively flat. In contrast, at a magnetic field frequency of 150Hz and a strength of 40mT, the weld surface is continuous but exhibits a significant amount of spatter on both the surface and the backside. The cross-sectional morphology shows slight concavity at the upper end and protrusion at the lower end. Therefore, in the full-penetration weldment state, a magnetic field frequency of 100Hz and a magnetic field strength of 20mT were selected, based on their effects on the weld morphology under different magnetic field frequencies and strengths.

3.2. Effect of alternating magnetic field on the morphology of partial-penetration welds

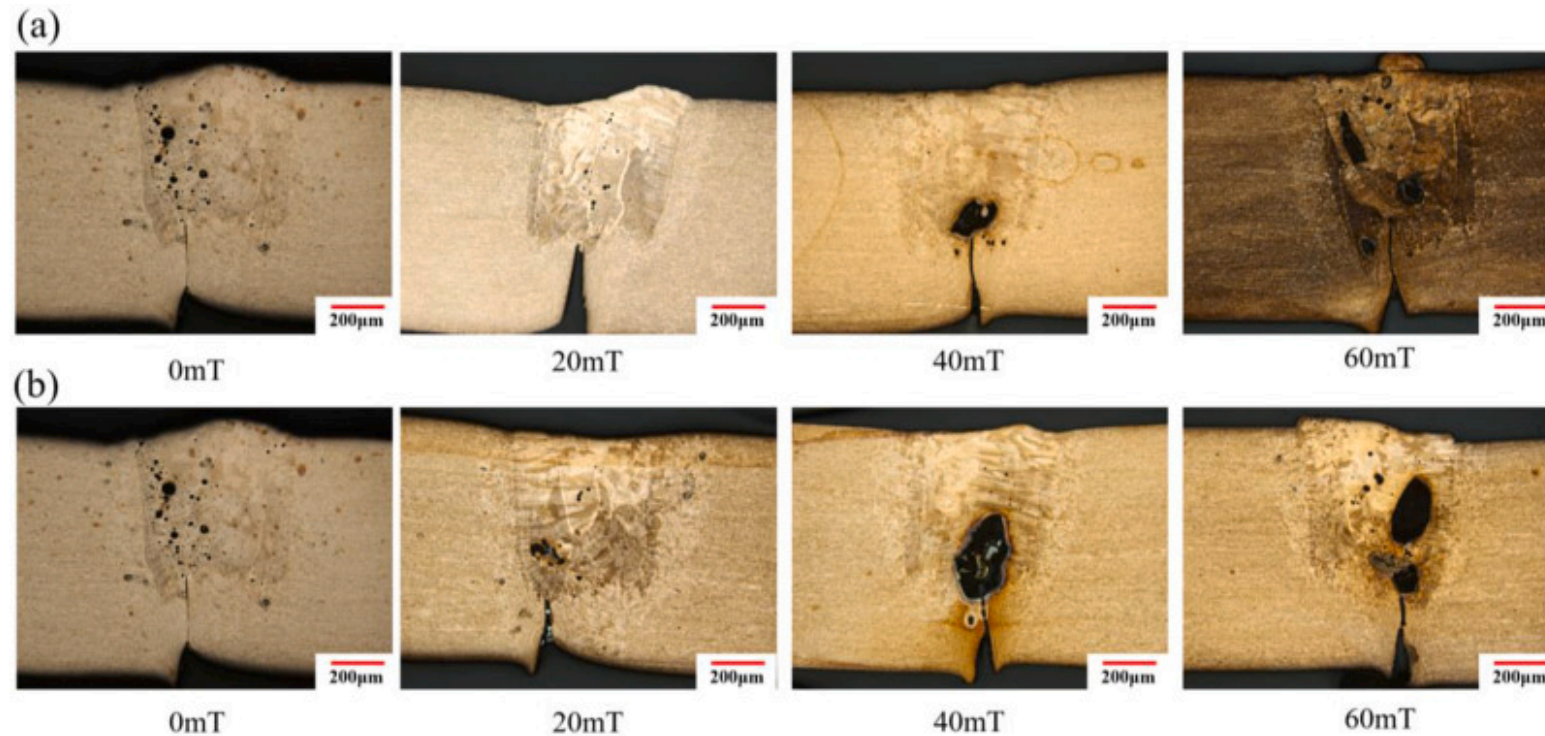
To investigate the influence of the alternating magnetic field on weld formation quality in both full- and partial-penetration T2 red copper joints, the effects of varying magnetic field frequencies and strengths on the weld seam surface and cross-sectional morphology of partial-penetration specimens were analyzed. These results were then compared with those of specimens without an applied magnetic field, as well as with full-penetration specimens subjected to the magnetic field. The findings are presented in [Fig. 6](#), [Fig. 7](#). As illustrated in [Fig. 6](#), in the absence of the alternating magnetic field, the weld seam surface of the partial-penetration specimen is highly uneven, with hump defects and metal spatter at the weld edge. The overall weld formation quality is poor. Additionally, as illustrated in [Fig. 7](#), a large number of small hole-type pores are present inside the weld in the absence of the magnetic field. This is primarily attributed to the failure of gas bubbles in the molten pool to escape in time during the welding process, leading to the entrapment of these bubbles within the weld area and the formation of porosity defects.



[Download: Download high-res image \(2MB\)](#)

[Download: Download full-size image](#)

Fig. 6. Weld surface morphologies of partial-penetration specimens under different alternating magnetic field frequencies and strengths: (a) 100Hz, (b) 150Hz.



[Download: Download high-res image \(951KB\)](#)

[Download: Download full-size image](#)

Fig. 7. Weld cross-section morphologies of partial-penetration specimens under different alternating magnetic field frequencies and strengths: (a)100Hz, (b)150Hz.

When the magnetic field frequency is 100Hz, as illustrated in Fig. 6, Fig. 7, with increasing magnetic field strength, the weld seam surface morphology of the partial-penetration specimen follows a trend similar to that of the full-penetrated specimen: it first improves and then worsens. At a magnetic field strength of 20mT, the weld seam surface becomes flatter and more continuous, with fine and clearly visible fish-scale ripples. Only localized raised defects are present, resulting in improved weld formation quality. Additionally, the porosity within the weld cross-section is significantly reduced compared to specimens without an applied magnetic field, indicating that the applied magnetic field effectively inhibits the formation of weld seam pores. However, when the magnetic field strength was increased to 40mT, the weld surface became discontinuous, with spatter and hump defects appearing. Although the number of small pores in the weld cross-section decreased, large process-induced pores emerged.

Overall, the weld formation quality declined compared to that at 20mT, though it remained better than that of the specimen without the applied magnetic field. When the magnetic field strength reaches 60mT, the weld surface exhibits irregular and uneven ripples, along with hump defects. The weld cross-section contains more impurities and process-induced pores, which are primarily concentrated in the center of the weld area, leading to poor weld formation quality. When the magnetic field frequency is increased to 150Hz, as seen in Fig. 6(b), the variation trend of the weld seam forming quality of the partial-penetration specimen is similar to that at 100Hz frequency. Within the magnetic field strength range of 20–60mT, the weld surface exhibits unevenness, with larger metal spatter particles and hump defects at the weld edges. The weld cross-section displays varying degrees of porosity and impurities. However, when subjected to a 20mT magnetic field, both the surface and cross-sectional morphology show the best overall weld formation, with the quality of the weld being superior to that under both the no-field condition and the other two magnetic field strengths.

Based on the above analysis, the frequency and strength of the alternating magnetic field exert a similar influence on weld formation quality in both full- and partial-penetration states. When the magnetic field frequency is constant, the weld seam surface and cross-sectional morphology initially improve and then worsen with increasing magnetic field strength. At a magnetic field frequency of 100Hz and a magnetic field strength of 20mT, the weld morphology is flat and continuous, with a fine and clearly visible fish-scale pattern. The cross-section contains only a small number of pores, indicating good weld formation quality. Similarly, when the magnetic field strength is constant, increasing the magnetic field frequency also causes the weld seam surface and cross-sectional morphology to initially improve and then deteriorate. This is primarily due to the influence of high-frequency electromagnetic forces, which can destabilize the molten pool, leading to turbulence that disrupts the relative stability of the molten pool flow, thereby reducing the quality of weld formation. Therefore, a magnetic field frequency of 100Hz and a magnetic field strength of 20mT were selected, considering their effects on the weld morphology and cross-sectional profile in the partial-penetration state, where the weldment has not fully melted through.

In summary, after applying 100Hz magnetic field frequency and 20mT magnetic field strength, whether the weld is penetrated or not, the weld morphology is continuous and fine fish scale ripples without obvious defects, and the weld shaping quality is superior. This can be attributed to the Lorentz force generated by the interaction between the alternating magnetic field and the current, which causes the molten metal flow to transition from a disordered to an ordered state. This promotes heat conduction, resulting in a more uniform temperature distribution within the melt pool, reducing the localized overheating caused by uneven cooling or temperature gradients. Consequently, the stability of the molten pool is enhanced, welding defects are minimized, and the overall weld formation quality is improved [29]. However, when a high alternating magnetic field strength is applied, the

Lorentz force in the molten pool becomes too strong, excessively stirring the molten metal. This disrupts the flow of the molten metal, reducing the stability of the melt pool. As a result, a large amount of molten metal is ejected, leading to metal spattering and a significant deterioration in the quality of the weld formation [30]. It has been shown that applying an optimal frequency and strength of the alternating magnetic field enhances the stability of the molten pool during welding, leading to significant improvements in weld formation quality. However, increasing the magnetic field strength beyond a certain threshold destabilizes the molten pool, resulting in defects such as spattering, humps, and porosity, which ultimately degrade the quality of the weld seam. It is noteworthy that under the influence of the alternating magnetic field, the weld formation quality in the full-penetration state is superior to that in the partial-penetration state. This can be attributed to the complete melting of the base material during full-penetration welding, which results in a smaller temperature gradient within the molten pool. As a result, the flow of the liquid metal becomes more uniform, helping to avoid defects caused by localized overheating and thereby improving the overall weld formation quality. Additionally, in the case of partial-penetration welds, the base material does not melt completely, leading to a significant temperature gradient between the molten pool and the base material. This results in relatively poor fluidity and heat transfer within the molten pool, thereby limiting the effectiveness of the alternating magnetic field in improving the weld formation quality.

4. Mathematical equations and computational models

Based on the optimized alternating magnetic field parameters and experimentally derived laser welding process conditions, a three-dimensional heat-fluid coupling numerical model is developed for T2 red copper laser welding assisted by an alternating magnetic field. This model aims to investigate the influence of the alternating magnetic field on the melt pool behavior of the specimen during both melt-through and non-melt-through welding processes.

4.1. Basic assumptions of the model

Due to the inherent complexity of the magnetic field-assisted laser welding process, which involves the coupled phenomena of heat conduction, fluid dynamics, and electromagnetism, the following basic assumptions must be adopted in the simulation and computation of the thermal-fluid coupled three-dimensional numerical model [31,32]: 1) The starting temperature of the base material is the same as the surrounding ambient temperature, which was 300K; 2) The liquid metal in the molten pool is an incompressible Newtonian fluid, and its motion is laminar flow; 3) The base material and the liquid metal in the molten pool are isotropic; 4) The heat generated during the formation of metal compounds in the molten pool is neglected; 5) The attenuation of

magnetic field strength during welding is neglected, and the base material is assumed to be exposed to a uniform alternating magnetic field; 6) The heat loss due to the shielding gas is neglected during the welding process.

4.2. Heat sources and control equations

T2 red copper laser welding is characterized by a high energy density, typical of deep fusion welding, where the heat flow density from the heat source affects not only the surface but also the depth of the workpiece. Considering the weld seam morphology in T2 red copper laser welding, the Gaussian rotating body heat source model was selected to analyze the temperature and flow fields in alternating magnetic field-assisted T2 red copper laser welding. The expression for this model [33] is as follows:

$$q(x, y, z) = \frac{9\eta Q}{\pi H r_0^2 (1 - e^{-3})} \exp \left[-\frac{9(x^2 + y^2)}{r_0^2 \ln(H/z)} \right] \quad (1)$$

Where η is the energy concentration coefficient of the Gaussian rotating body heat source; Q is the effective thermal power of the laser beam; H is the height of the laser heat source; and r_0 is the radius of the laser heat source's focal spot.

Based on the aforementioned assumptions, the key characteristics of the weld melt pool, heat transfer, and fluid flow are described using the corresponding governing equations [34], which primarily include:

Fluid continuity equation:

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0 \quad (2)$$

Energy conservation equation:

$$\frac{\partial(\rho T)}{\partial t} + \frac{\partial(\rho u T)}{\partial x} + \frac{\partial(\rho v T)}{\partial y} + \frac{\partial(\rho w T)}{\partial z} = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + S_H \quad (3)$$

Momentum conservation equation:

$$\begin{aligned} \frac{\partial(\rho u)}{\partial t} - \frac{\partial(\rho u_0 u)}{\partial x} + \frac{\partial(\rho u u)}{\partial x} + \frac{\partial(\rho v u)}{\partial y} + \frac{\partial(\rho w u)}{\partial z} &= \frac{\partial}{\partial x} \left(\mu \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu \frac{\partial u}{\partial y} \right) + \\ \frac{\partial}{\partial z} \left(\mu \frac{\partial u}{\partial z} \right) - \frac{\partial P}{\partial x} + S_x & \end{aligned} \quad (4)$$

$$\frac{\partial(\rho v)}{\partial t} - \frac{\partial(\rho u_0 v)}{\partial x} + \frac{\partial(\rho u v)}{\partial x} + \frac{\partial(\rho v v)}{\partial y} + \frac{\partial(\rho w v)}{\partial z} = \frac{\partial}{\partial x} \left(\mu \frac{\partial v}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu \frac{\partial v}{\partial y} \right) + \frac{\partial}{\partial z} \left(\mu \frac{\partial v}{\partial z} \right) - \frac{\partial P}{\partial x} + S_y \quad (5)$$

$$\frac{\partial(\rho w)}{\partial t} - \frac{\partial(\rho u_0 w)}{\partial x} + \frac{\partial(\rho u w)}{\partial x} + \frac{\partial(\rho v w)}{\partial y} + \frac{\partial(\rho w w)}{\partial z} = \frac{\partial}{\partial x} \left(\mu \frac{\partial w}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu \frac{\partial w}{\partial y} \right) + \frac{\partial}{\partial z} \left(\mu \frac{\partial w}{\partial z} \right) - \frac{\partial P}{\partial z} + S_z \quad (6)$$

where ρ is the density; u , v , and w are the velocities of the molten metal in the molten pool along the x , y , and z directions, respectively; T is the temperature; t is the time; k is the thermal conductivity; S_H is the source term of the energy conservation equation; S_x , S_y , and S_z , are the source terms of the momentum conservation equations in the x , y , and z directions, respectively; u_0 is the welding speed; μ is the dynamic viscosity of the molten metal; P is the pressure.

The Volume-of-Fluid (VOF) method was employed to track the evolution of the molten metal interface within the molten pool, including both the surface of the pool and the walls of the keyhole. Its VOF control equation is [35]:

$$\frac{\partial F}{\partial t} + \nabla \cdot (\vec{v} \cdot F) = 0 \quad (7)$$

where $\vec{v}$ is denoted as the velocity vector of the fluid flow in the melt pool in the x , y , and z directions; F is denoted as the volume of the fluid in the melt pool in the x , y , and z directions.

In the alternating magnetic field-assisted laser scanning welding process of T2 red copper, the temperature gradient between the solid and liquid phases in the molten pool induces a voltage difference, generating a thermal current, which is referred to as the Seebeck effect [36]. Meanwhile, the alternating magnetic field-assisted fluid flow is realized by adding volumetric Lorentz force and Joule heat to the source terms of the momentum and energy conservation equations, respectively, which are governed by the electromagnetic field control equations [37,38]:

$$-\sigma_e (\nabla \varphi_e + S \nabla T) = J_E \quad (8)$$

$$\nabla \cdot J_E = 0 \quad (9)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (10)$$

$$\nabla \times \mathbf{B} = \mu_m \mathbf{J}_E \quad (11)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (12)$$

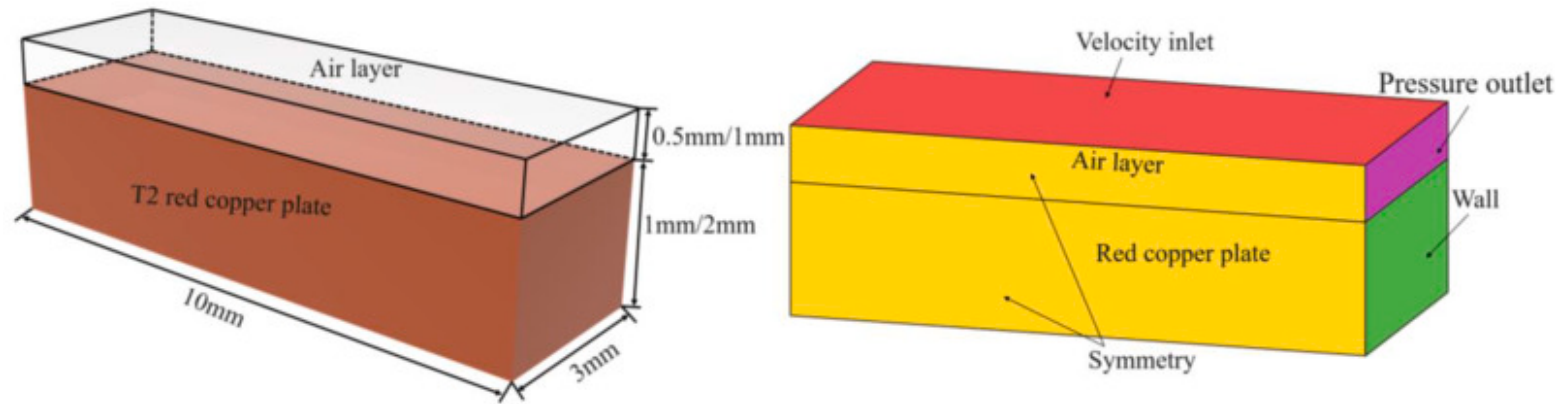
$$\mathbf{B} = \nabla \times \mathbf{A} \quad (13)$$

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t} \quad (14)$$

Where σ_e is the material conductivity; ∇ is the Hamiltonian operator; S is the Seebeck coefficient; T is the temperature; $\mathbf{J}_E$ is the eddy current density vector; $\mathbf{B}$ is the magnetic induction vector; μ_m is the magnetic permeability in vacuum; $\mathbf{E}$ is the electric field strength vector; $\mathbf{A}$ is the magnetic vector potential; V is the electric potential.

4.3. Modeling and boundary conditions

The simulated geometric model consists of an air layer at the top and T2 red copper at the bottom, with the thickness of the air layer being half that of the T2 red copper. The total dimensions and boundary conditions of the simulation model for T2 red copper in both the full- and partial-penetration states are illustrated in Fig. 8. For the full-penetration specimen, the air layer and specimen thicknesses are 0.5mm and 1 mm, respectively, while for the partial-penetration specimen, the air layer and specimen thicknesses are 1 mm and 2mm, respectively. The structured meshing tool ICEM CFD was employed to mesh the workpieces, with an encrypted mesh applied to the T2 red copper plate and a sparse mesh used for the air layer, the quality of all mesh cells was maintained above 0.95. After importing the geometric model into FLUENT simulation software, the heat source is advanced along the welding direction at a uniform speed by writing a Gaussian body heat source modeling program, followed by applying an alternating magnetic field on the workpiece and defining the corresponding boundary conditions. According to the weld morphology analysis in Section III, it can be seen that when the magnetic field frequency is 100Hz and the magnetic field strength is 20mT, the forming quality of the weldment is better in both the full- and partial-penetration states. Therefore, 100Hz magnetic field frequency and 20mT magnetic field strength are chosen to analyze the numerical simulation of the melting pool of the full- and partial-penetration weld.



[Download: Download high-res image \(335KB\)](#)

[Download: Download full-size image](#)

Fig. 8. Simulation geometric model and boundary conditions.

4.4. Simulation material parameters

The material studied in this paper is T2 red copper, and its main physical performance parameters are listed in [Table 3](#).

Table 3. Physical performance parameters of T2 red copper.

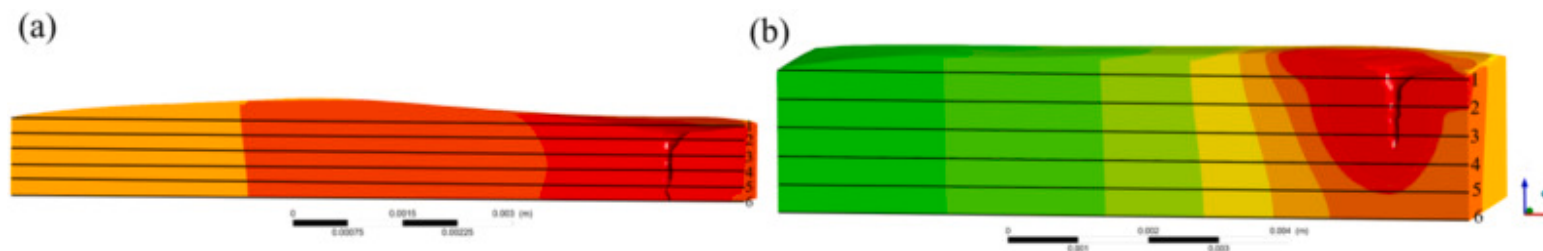
Physical parameter	Symbol	Unit	Numerical value
Material density	ρ	g/cm^3	8.96
Viscosity	μ	$\text{kg}/(\text{m}\cdot\text{s})$	3.41×10^{-3}
Specific heat capacity	C_p	$\text{J}/(\text{g}\cdot\text{K})$	3.86×10^5
Latent heat of fusion	L_y	J/Kg	2.07×10^5
Latent heat of vapor	L_m	J/Kg	2.26×10^6
Heat conductivity	λ	$\text{W}/(\text{m}\cdot\text{K})$	401

Physical parameter	Symbol	Unit	Numerical value
Solid temperature	T_S	K	710
Liquid temperature	T_L	K	1083.4
Boiling temperature	T_g	K	2562
Coefficient of thermal expansion	β	$\times 10^{-6} \text{ } ^\circ\text{C}^{-1}$	16.6
Surface tension coefficient	σ	N/m	1.103
Temperature coefficient of surface tension	$d\sigma/dT$	N/(m·K)	-1.72×10^{-6}

5. Simulation results and analysis

5.1. Effect of alternating magnetic field on the temperature field of full- and partial-penetration weld pools

To analyze the effect of the alternating magnetic field on the temperature of the weld pool in both full- and partial-penetration states, the molten pool temperature at the 220ms time node is selected for analysis. This time node is chosen due to the relatively small temperature fluctuations in the weldment, indicating a more stable molten pool state. In the longitudinal (X-Z) section of the full - and partial-penetration welding pool, 6 lines are taken from the upper surface of the pool to the bottom, and are named lines 1 to 6 in turn, as shown in Fig. 9. The temperature of molten pool in the full- and partial-penetration state of these 6 lines was analyzed.

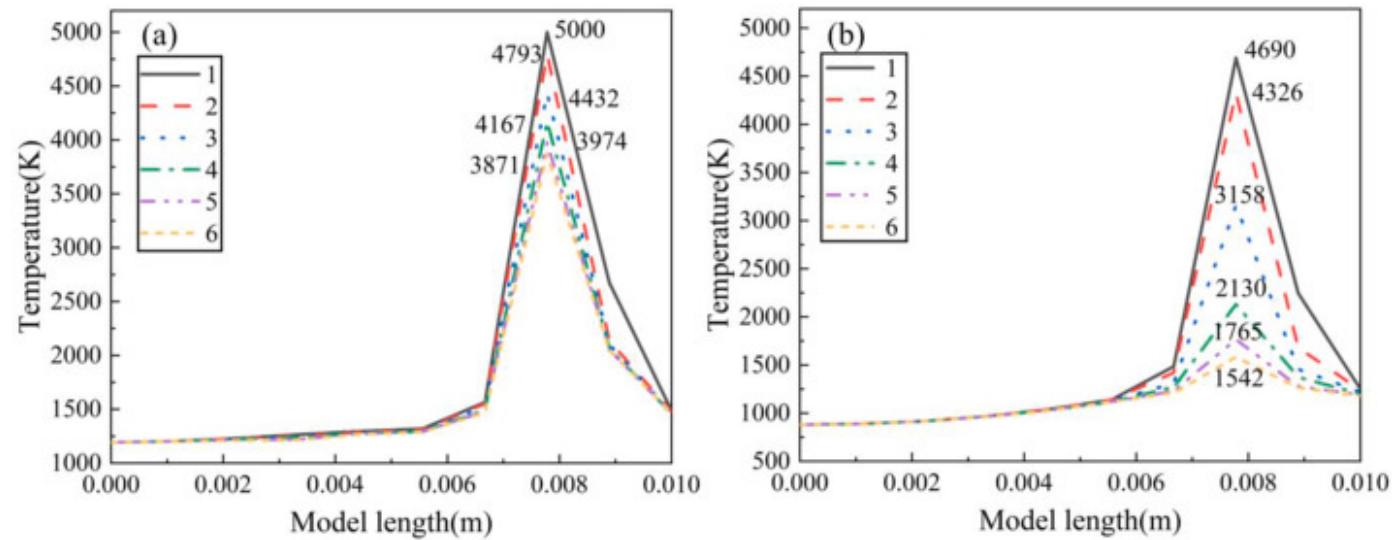


[Download: Download high-res image \(246KB\)](#)

[Download: Download full-size image](#)

Fig. 9. Schematic diagram of the position of the lines of specimens: (a) full-penetration specimen, (b) partial-penetration specimen.

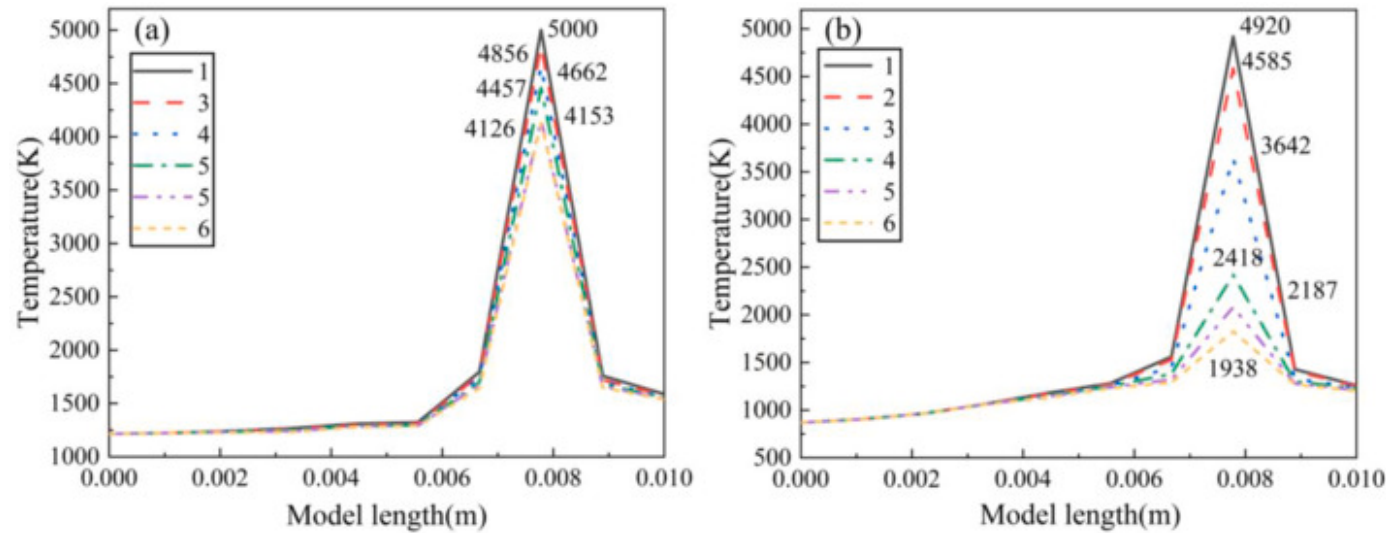
Fig. 10, Fig. 11 display the temperature profiles of the full- and partial-penetration weld pools, with and without the application of an alternating magnetic field, during the 220ms stabilization phase. As observed in the figures, the temperature variation in the weld pool, with and without the applied magnetic field, follows a similar trend. The temperature increases as the distance from the pool surface decreases, reaching its highest values near the center of the pool. In the absence of a magnetic field, the maximum and minimum temperatures of the molten pool for the full- and partial-penetration weldments were 5000K and 4690K, and 3871K and 1542K, respectively; the corresponding temperature differences in the molten pool were 1129K and 3148K, respectively; after the application of magnetic field strength, the maximum and minimum temperatures of the molten pool for the full- and partial-penetration weldments were 5000K and 4920K, and 4126K and 1938K, respectively; the corresponding temperature differences in the molten pool were 874K and 2982K, respectively. It is evident that the temperature difference in the molten pool of the full-penetration weldments is significantly lower than that of the partial-penetration weldments, suggesting a more uniform internal temperature distribution in the molten pool of the full-penetration weldments, which helps to reduce the overall temperature gradient. Additionally, the temperature difference in the molten pool, both in the full- and partial-penetration states, is reduced compared to the case without the magnetic field. Specifically, the temperature difference decreases by 22.59% in the full-penetration state and 5.27% in the partial-penetration state, with the reduction being more pronounced in the melted state under the influence of the alternating magnetic field. The reason is that under the action of the alternating magnetic field, the melt pool will produce an induced current, which triggers the electromagnetic stirring effect, changing the flow pattern of the molten pool, and prompting the formation of a strong rotational flow in the melt pool, which helps to enhance the heat convection inside the molten pool, so that the heat is more evenly distributed. In full-penetration welding, the laser energy is sufficient to fully penetrate the workpiece, creating a molten pool that extends through the entire thickness of the material. Applying an alternating magnetic field further enhances the overall fluidity of the molten pool, particularly at its bottom, where the intensified circulation of liquid metal significantly improves convective heat transfer. As a result, the heat is more evenly distributed within the molten pool, reducing the temperature gradient and promoting uniform solidification and cooling of the weld. In contrast, in unfused welding, the improvement in the temperature gradient is relatively limited, as the bottom of the molten pool remains solid, restricting the flow of liquid metal and hindering the electromagnetic stirring effect of the alternating magnetic field, which leads to localized heat accumulation.



[Download: Download high-res image \(415KB\)](#)

[Download: Download full-size image](#)

Fig. 10. Temperature distribution curves at different locations in the melt pool of full- and partial-penetration specimens without the application of an alternating magnetic field: (a) full-penetration, (b) partial-penetration.



[Download: Download high-res image \(390KB\)](#)

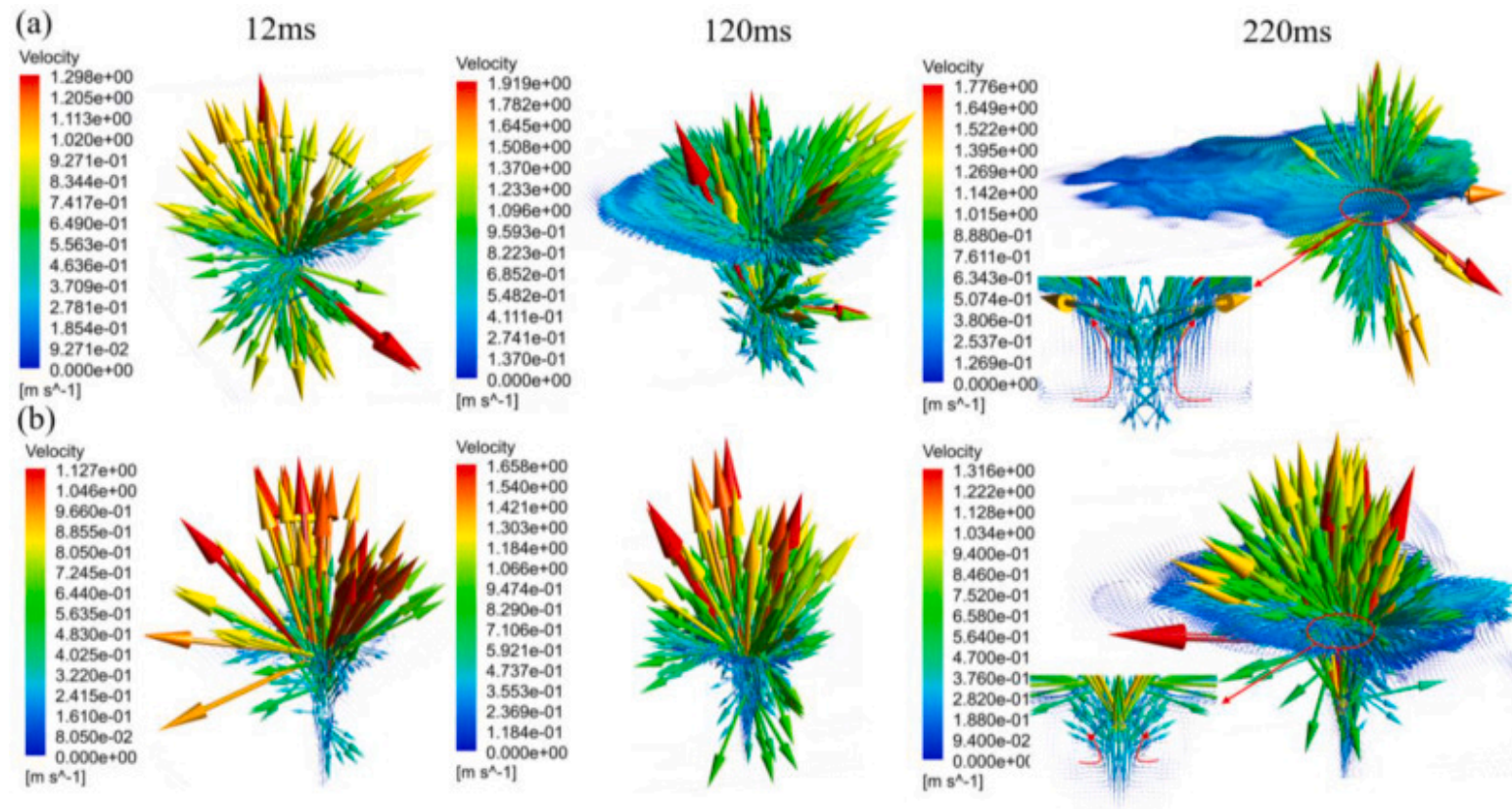
[Download: Download full-size image](#)

Fig. 11. Temperature distribution curves of the melting pool of the full- and partial-penetration specimens at different locations under the influence of an alternating magnetic field:(a) full-penetration, (b) partial-penetration.

Furthermore, the research team previously found that during the laser welding of T2 red copper, the violent eruption of the metal vapor and plasma plume significantly scatters and absorbs the laser beam energy, which in turn affects the stability of the welding process and leads to energy loss [39]. Upon the application of a specific magnetic field strength, the electromagnetic effects within the molten pool cause the metal vapor and plasma mixed plume to contract. This contraction significantly weakens the shielding effect of the laser-induced plume, thereby reducing laser energy loss. As a result, the stability of the molten pool is markedly improved, leading to a more efficient thermal input of laser energy. Therefore, the application of an appropriate magnetic field strength effectively reduces the shielding effect caused by metal vapor and plasma, enhancing the utilization of laser energy within the molten pool. This, in turn, increases the overall temperature of the molten pool, helping to reduce the temperature gradient. Consequently, the cooling rate within the molten pool becomes more uniform, preventing premature solidification and maintaining the fluidity of the molten pool, thereby improving the stability and efficiency of the welding process.

5.2. Effect of alternating magnetic field on the flow field of full- and partial-penetration weld pools

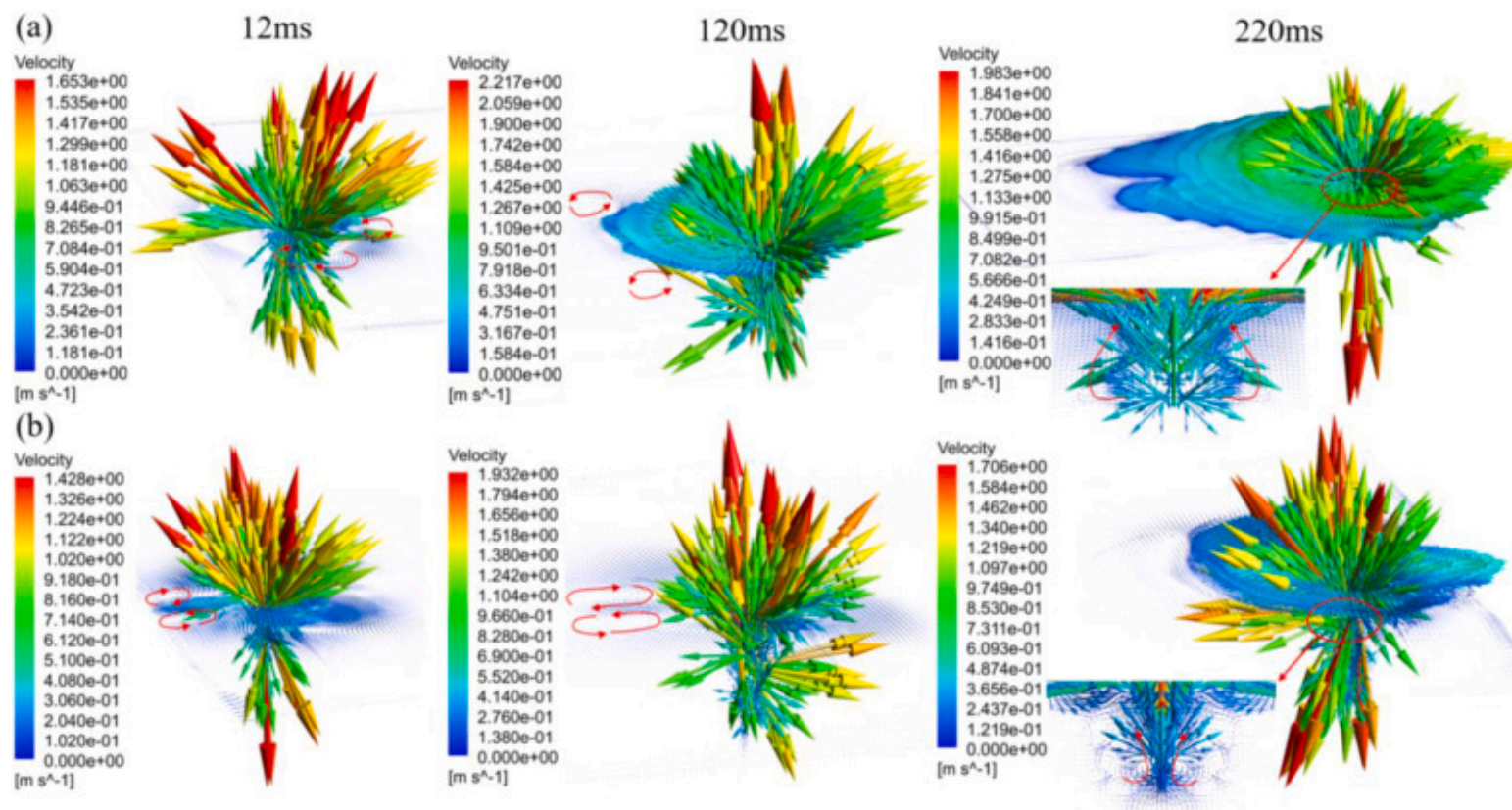
Fig. 12, Fig. 13 show vector plots of cross-sectional flow velocity distributions in the melt pool for full- and partial-penetration specimens at different stages with and without an applied magnetic field. As illustrated in the figures, regardless of the presence or absence of the applied alternating magnetic field, the fluid in the molten pool flows radially outward from the center to the periphery in both full- and partial-penetration weldments. The flow velocity is highest at the center and decreases toward the edges. This phenomenon is attributed to the increased friction between the molten metal and the solid substrate at the molten pool boundary, which leads to a higher viscous force in the boundary layer. Consequently, the fluid flow velocity near the boundary is significantly lower than at the center, as the combined effect of viscous forces and friction suppresses the flow at the edges. Regardless of whether the weld is penetrated or not, when the magnetic field is not applied, because the temperature coefficient of surface tension of the molten pool of red copper is negative and decreases gradually with increasing temperature, the molten metal flows from the high-temperature region to the low-temperature region, resulting in the formation of Marangoni convection from the center of the molten pool to the surrounding area. During the welding process, the heat source is concentrated in the center of the molten pool, resulting in a large temperature gradient, so that the density of the molten metal in the central region decreases and rises, while the surrounding colder metal sinks, resulting in the formation of a more intense circulating convection, which promotes the expansion of the fluid in all directions. The molten pool fluid is driven by Marangoni convection and buoyancy forces. Molten metal near the keyhole sides of the pool flows upward from the pool's bottom, while molten metal near the solid-liquid interface at the front and rear ends of the pool is driven downward by gravity and finally converges to form a vortex at the bottom of the pool.



[Download: Download high-res image \(1MB\)](#)

[Download: Download full-size image](#)

Fig. 12. Vector plots of the flow velocity distribution at different stages of the melt pool of the full - and partial-penetration specimens without applied alternating magnetic field: (a) full-penetration, (b) partial-penetration.



[Download: Download high-res image \(1MB\)](#)

[Download: Download full-size image](#)

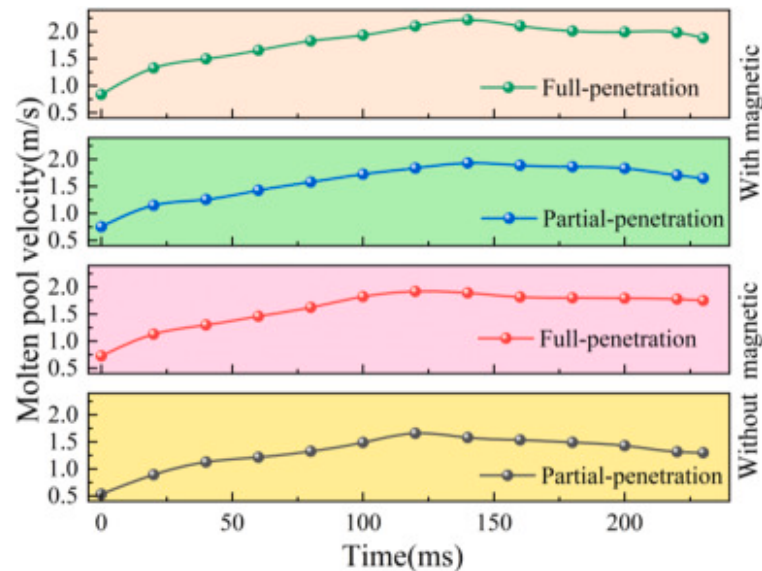
Fig. 13. Vector plots of flow velocity distribution in the melt pool of full- and partial-penetration welds at different stages under the influence of an applied alternating magnetic field: (a) full-penetration, (b) partial-penetration.

After the application of the alternating magnetic field, the fluid in the melt pool continues to flow from the center to the surrounding area. However, the two symmetric counter-rotating vortices within the melt pool become more pronounced compared to the case without the magnetic field. These vortices enhance the homogeneous flow of the molten metal, promoting more efficient heat exchange between the interior and exterior of the melt pool. This reduction in the temperature gradient facilitates improved crystal growth and further enhances the solidification process. The reason for this is that when an alternating magnetic field is applied to the red copper melt pool, the high electrical conductivity of red copper, combined with

the interaction between the magnetic field and the laser heat source, induces a current within the melt pool. The interaction between this induced current and the magnetic field generates the electromagnetic effect, particularly the Lorentz force, which creates a significant driving force on the fluid in the melt pool, resulting in the formation of helical flow and enhancing convection within the molten pool. The reason for this is that when an alternating magnetic field is applied to the red copper melt pool, due to the high electrical conductivity of T2 red copper, combined with the interaction between the magnetic field and the laser heat source, induces a current within the melt pool. The interaction between this induced current and the magnetic field generates the electromagnetic effect, particularly the Lorentz force in the electromagnetic effect, which creates a significant driving force on the fluid in the melt pool, resulting in the formation of helical flow and enhancing convection within the molten pool. It is noteworthy that after the application of the alternating magnetic field, the eddy currents at the bottom of the molten pool in the full-penetration fusion state are more intense and have a wider influence compared to those in the partial-penetration state. This is because full-penetration welding eliminates the obstruction to liquid metal flow caused by the solid metal at the bottom of the workpiece, thereby promoting the efficient diffusion of liquid metal within the molten pool. As a result, the full-penetration pool flow range is significantly increased in the molten state compared to the partial-penetration state.

Fig. 14 illustrates the maximum flow velocity of the molten pool of the full- and partial-penetration specimens with and without the applied magnetic field at different stages. From the figure, regardless of whether a magnetic field is applied, the maximum flow velocity of the molten pool for both the full- and partial-penetration specimens initially increases and then decreases. Additionally, the maximum flow rate in the full-penetration state is higher than that in the partial-penetration state. This is primarily because the temperature distribution in the molten pool of the full-penetration state is more uniform, the heat conduction efficiency is higher, and the absence of a solid interface at the bottom allows freer metal flow and enhanced convection, thereby increasing the overall flow velocity. In contrast, in the partial-penetration state, the bottom of the molten pool remains solid, and the presence of solid-liquid interfaces restricts fluid flow and weakens convection, leading to a lower overall flow velocity. The maximum flow velocity of the molten pool significantly increased in both full- and partial-penetration specimens after applying the alternating magnetic field. Specifically, the maximum flow velocities of the full- and partial-penetration specimens reached 2.217 m/s and 1.932 m/s, respectively, representing increases of 15.53% and 16.53% compared to those without the magnetic field. This phenomenon indicates that in the alternating magnetic field-assisted laser welding of T2 red copper, the molten pool flow characteristics are influenced by the combined effects of the temperature gradient, Marangoni convection, and electromagnetic force. The flow evolution can be divided into three phases: (1) Initial phase: The laser beam

rapidly heats the material, forming a steep temperature gradient within the molten pool, which induces strong Marangoni convection. Simultaneously, the Lorentz force generated by the alternating magnetic field drives the motion of the liquid metal, further enhancing the molten pool flow and gradually increasing the flow velocity. (2) Growth stage: As the volume and depth of the keyhole increase, the molten pool maintains a large temperature gradient. Thermal convection and electromagnetic stirring work synergistically to further promote liquid metal flow within the molten pool, causing the flow velocity to reach its peak. (3) Stabilization and cooling phase: As the laser energy input decreases, the temperature of the molten pool gradually drops, leading to an increase in fluid viscosity and, consequently, higher flow resistance of the liquid metal. Meanwhile, the reduced temperature weakens the electromagnetic stirring effect, diminishing the Lorentz force acting on the molten pool. This results in a decrease in the flow rate, ultimately causing the molten pool flow to slow down and solidify. As a result, the molten pool flow velocity exhibits a characteristic of increasing and then decreasing, reflecting the complex coupling between laser energy input, heat transfer, fluid dynamics, and electromagnetic effects.



[Download: Download high-res image \(316KB\)](#)

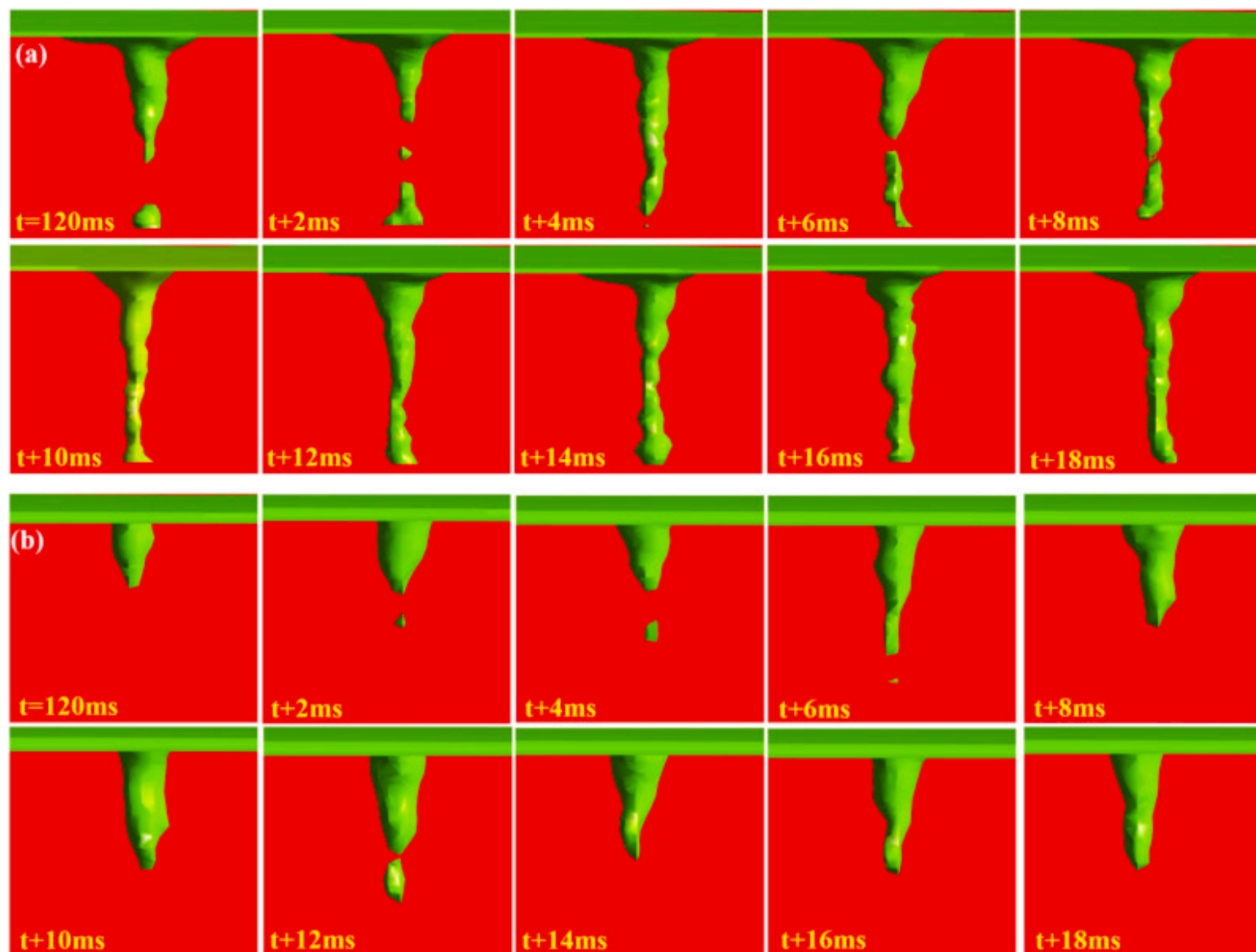
[Download: Download full-size image](#)

Fig. 14. Melt pool flow velocity curves of full- and partial-penetration specimens at different stages with or without an alternating magnetic field.

It is worth noting that the alternating magnetic field significantly improves the flow velocity of the full-penetration specimens compared to the partial-penetration specimens. This improvement is attributed to the fact that full-penetration welding completely melts the base material, removing the restriction on molten pool flow caused by the solid-state interface, and thereby enhancing the mobility of the liquid metal. At the same time, the synergistic effect of electromagnetic stirring reduces the viscous resistance between the molten pool surface and the underlying liquid, thereby enhancing the maximum flow velocity of the molten pool.

5.3. Effect of applied alternating magnetic field on keyhole stability in full- and partial-penetration melt pools

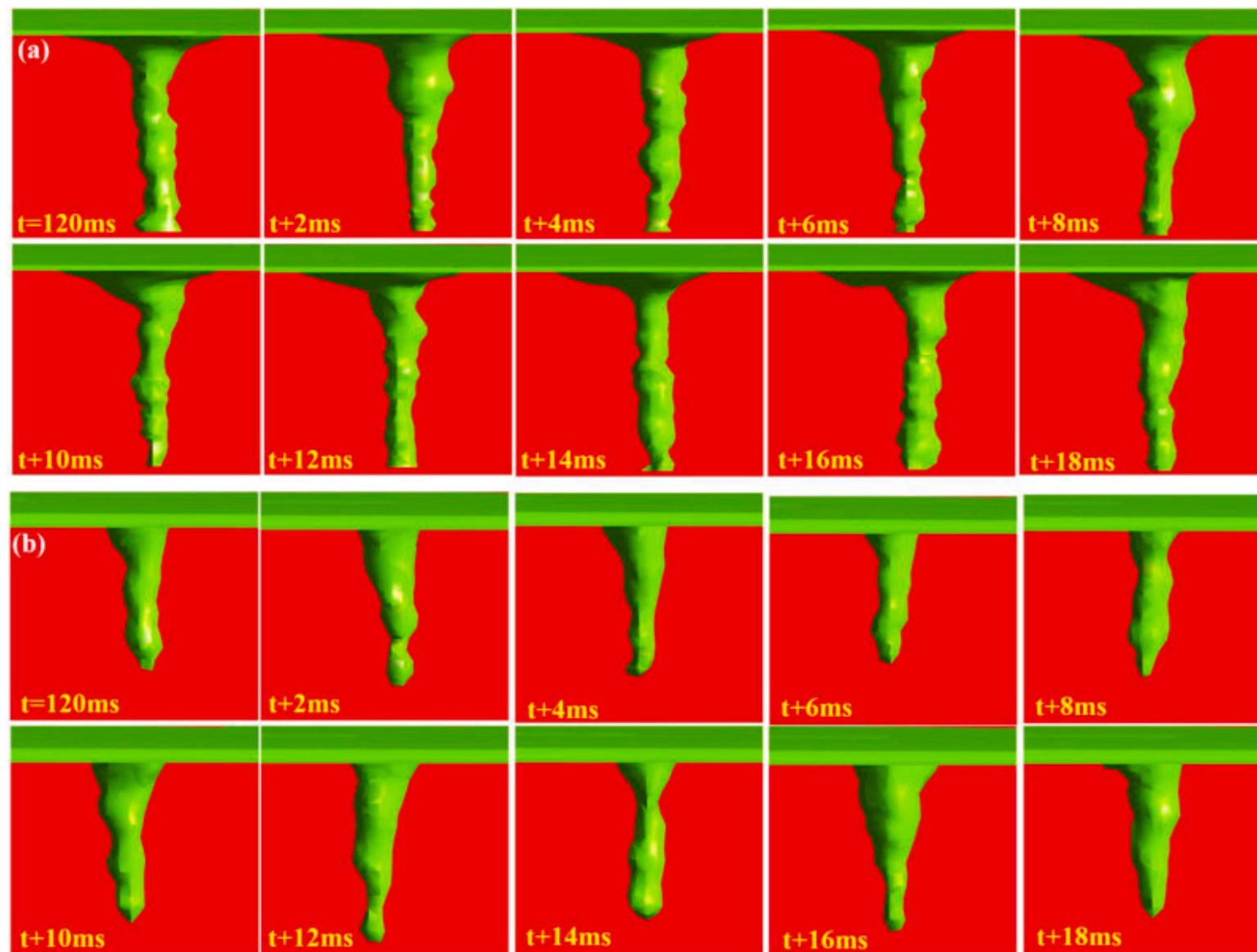
[Fig. 15](#), [Fig. 16](#) illustrate the evolution of the three-dimensional keyhole morphology in both full- and partial-penetration specimens, with and without the application of an alternating magnetic field, over a certain period. As shown in the figure, the keyholes of both full- and partial-penetration specimens under the unapplied magnetic field collapsed several times throughout the interception time period, and there were hump defects on the walls of the keyholes, which led to irregular deformations. The reason for this is that during the laser welding process, the laser energy induces material evaporation, producing a high-temperature metal vapor; that vapor generates recoil pressure within the keyhole [40]. If the recoil pressure is unbalanced, it can destabilize the keyhole, leading to shape oscillations or premature closure; conversely, excessive recoil pressure may cause localized keyhole collapse. In addition, full-penetration welding can make the molten pool completely melted, liquid metal mobility is better, and it can accelerate the escape of gas and steam, reducing the pressure fluctuations and gas accumulation inside the molten pool, thus forming a more regular keyhole; while in the partial-penetration state, due to the laser energy does not make the material internal to reach the full melting and vaporization, the instability of the keyhole morphology is more significant, and it is more likely to lead to the irregular deformation of the shape of keyholes or closure.



Download: [Download high-res image \(836KB\)](#)

Download: [Download full-size image](#)

Fig. 15. Evolution of keyhole morphology in full- and partial-penetration specimens without the application of an alternating magnetic field: (a) full-penetration, (b) partial-penetration.



[Download: Download high-res image \(1MB\)](#)

[Download: Download full-size image](#)

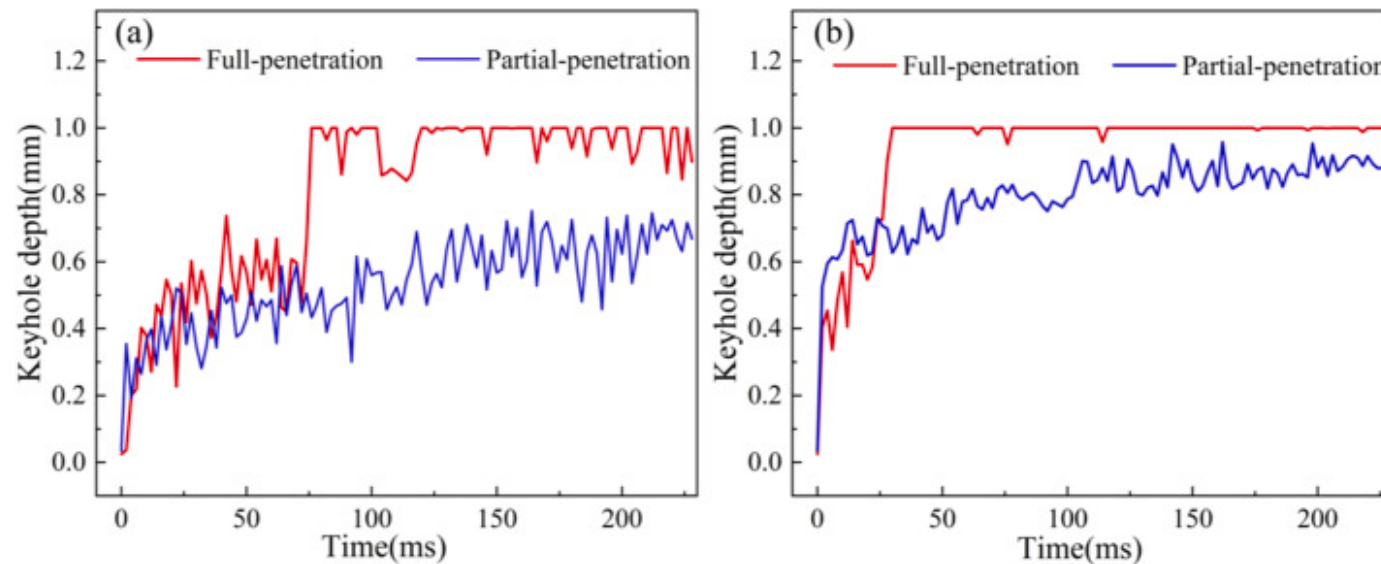
Fig. 16. Evolution of keyhole morphology in full- and partial-penetration specimens under the influence of an alternating magnetic field: (a) full-penetration, (b) partial-penetration.

According to the previous analysis of the temperature and flow characteristics within the molten pool, the absence of an applied magnetic field results in a larger temperature gradient and more unstable convection within the molten pool. This instability, coupled with variations in surface tension, causes disordered liquid metal flow, which compromises the stability of the keyhole and can lead to keyhole deformation and, in severe cases, keyhole collapse. Applying an alternating magnetic field induces an electromagnetic stirring effect within the molten pool, enhancing the convection of the liquid metal and promoting a more stable and orderly flow. This improved flow facilitates uniform heat distribution, effectively minimizing localized overheating or overcooling. As a result, the risk of keyhole deformation or collapse due to excessive temperature is significantly reduced. Simultaneously, the addition of magnetic field forms a new pressure balance on the keyhole wall, reduces the instability of the keyhole shape caused by excessive local pressure, helps to keep the keyhole opening and make the keyhole shape more regular.

In the full-penetration state, the alternating magnetic field can improve the fluidity of the melt pool, uniform heat distribution, enhance the vapor pressure, prevent excessive loss of liquid metal, and reduce the eddy current intensity at the bottom of the keyhole to maintain the stability of the depth and shape of the keyhole; whereas, in the partial-penetration state, the alternating magnetic field will electromagnetically drive the internal fluid to generate eddy currents or circulating flow, which promotes the transfer of laser energy to a deeper layer, improves the vaporization effect as well as reduces the closure of the air holes, and thus significantly increases the depth of the keyholes and the stability of the morphology. Consequently, the stability of the keyhole is notably enhanced in both full- and partial-penetration states. It is noteworthy that the alternating magnetic field has a more significant effect on the maintenance of the keyhole morphology in the full-penetration state, which is due to the fact that the full-penetration welding can make the liquid metal flow more fully, and the addition of the magnetic field further enhances the convection and fluidity of the molten pool, which optimizes the keyhole morphology and avoids irregular deformations. In contrast, in partial-penetration welding, the depth and mobility of the molten pool are limited, so that gas and vapor are more likely to accumulate in the molten pool, resulting in an unstable morphology of the keyhole, which is prone to irregular deformation. As a result, full-penetration welding provides better control over the morphology of the keyhole compared to partial-penetration welding, thus ensuring consistent weld quality. To further elucidate the effect of the alternating magnetic field on keyhole depth in both the full- and partial-penetration states, the variation in keyhole depth throughout the welding process is subsequently analyzed.

Fig. 17 show to curves of keyhole depth in the melt pool as a function of time for full- and partial-penetration states, with and without the application of an alternating magnetic field. By analyzing the variation curves of keyhole depth, the effect of the magnetic field on keyhole stability in both full- and partial-penetration welding states throughout the welding process is

investigated. As shown in the figure, regardless of whether the alternating magnetic field is applied, the keyhole depth in both the full- and partial-penetration states increases rapidly in the early stages of welding, exhibits significant oscillations during the middle stage, and tends to stabilize in the late stage. The application of an alternating magnetic field shortens the fluctuation time of the keyhole depth in the middle stage, indicating that the introduction of the magnetic field helps to maintain the stability of the keyhole's depth and morphology to some extent. This is because electromagnetic stirring under an alternating magnetic field significantly enhances flow uniformity and temperature distribution within the molten pool, thereby balancing the non-uniform recoil pressure induced by laser evaporation and stabilizing the keyhole depth and morphology. The high electrical conductivity of T2 red copper enables the laser heat source to interact with the magnetic field, inducing electromagnetic effects and generating Lorentz forces. These forces not only drive the movement of liquid metal in a specific direction but also stabilize local temperature and flow rate fluctuations, effectively preventing recoil pressure instability caused by temperature non-uniformity. Meanwhile, the uniform convection induced by the alternating magnetic field facilitates the rapid diffusion of local high pressure, mitigating pressure shocks caused by metal vapor during laser evaporation; it reduces the surface temperature gradient of the molten pool, enhancing Marangoni convection and further suppressing violent flow induced by surface tension inhomogeneity. Therefore, under the synergistic influence of these effects, heat and fluid distribution in the molten pool become more uniform, effectively suppressing keyhole oscillation and irregular deformation, thereby enhancing the overall weld quality and consistency.



[Download: Download high-res image \(400KB\)](#)

[Download: Download full-size image](#)

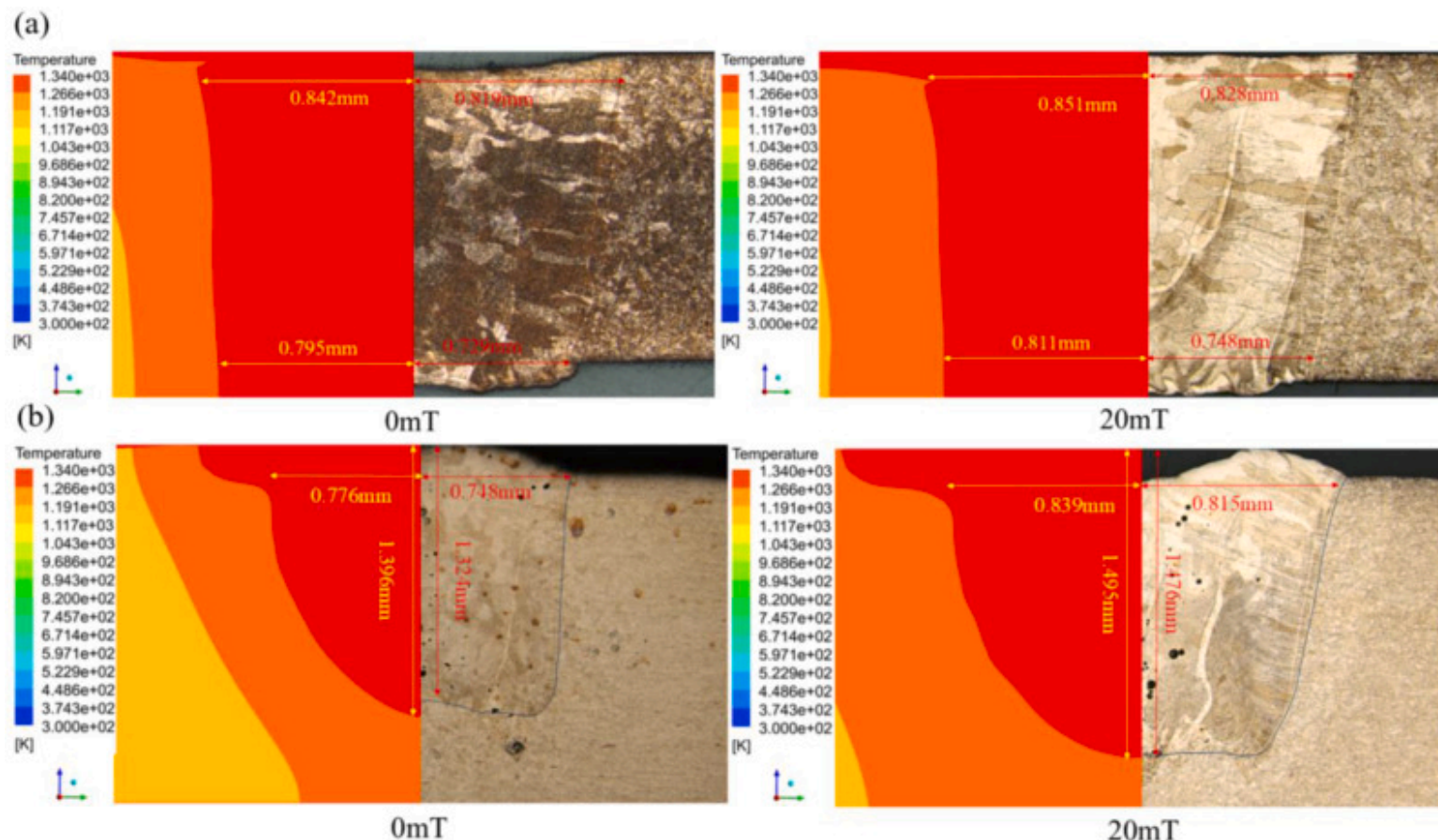
Fig. 17. Variation curve of keyhole depth in the molten pool for full- and partial-penetration specimens with or without an alternating magnetic field: (a) without magnetic field, (b) with applied magnetic field.

Furthermore, in the full-penetration state, the keyhole is deeper and more stable. In contrast, in the partial-penetration state, the keyhole is shallower. Although electromagnetic forces reduce depth fluctuations, the overall stability remains lower than that of the full-penetration state, and the keyhole depth is still smaller. Comparative analyses indicate that the effects of alternating magnetic fields on keyhole stability differ between full- and partial-penetration states. In the full-penetration state, the magnetic field has a more pronounced stabilizing effect on keyhole depth and morphology, helping to maintain stability. However, in the partial-penetration state, while the magnetic field enhances keyhole stability to some extent, it does not significantly reduce depth fluctuations. Therefore, the alternating magnetic field has a more significant effect on enhancing keyhole stability in the full-penetration state, while its impact on the partial-penetration state is relatively limited.

6. Experimental results analysis and simulation verification

6.1. Experimental verification of melt pool morphology

To validate the accuracy of the numerical model and the simulation results, an alternating magnetic field-assisted laser welding experiment was conducted on T2 red copper using the same laser process parameters as those employed in the numerical simulation. The cross-sectional morphology of the full- and partial-penetration welded specimens under the influence of the alternating magnetic field was compared with the fusion line profiles obtained from the simulation calculations, as shown in [Fig. 18](#). As shown in the figure, after the application of an alternating magnetic field, the upper and lower melting width of the weld in the state of full-penetration are 1.656mm and 1.496mm, respectively. Compared with the upper melting width (1.702mm) and lower melting width (1.622mm) obtained by simulation, the error is about 2.70% and 7.77%. The weld width and depth were 1.63mm and 1.476mm respectively in the partial-penetration state, and the error was about 2.86 % and 1.27 % compared with the simulated width (1.678mm) and depth (1.495mm).



[Download: Download high-res image \(1MB\)](#)

[Download: Download full-size image](#)

Fig. 18. Comparison of simulation results and experimental results of the contours of full- and partial-penetration joints with or without an alternating magnetic field: (a) full-penetration, (b) partial-penetration.

Comparative analysis shows that the welding experiments and simulation calculations obtained by the weld cross-section fusion morphology of the basic consistency, but the overall existence of some minor differences, is mainly reflected in the following two aspects. For one thing, there is a discrepancy between the numerical model and the actual material properties. (1) In the simulation, it is assumed that the liquid metal in the molten pool behaves as an incompressible, laminar, Newtonian fluid with isotropic properties. However, the actual material properties in welding are anisotropic, and this discrepancy may lead to

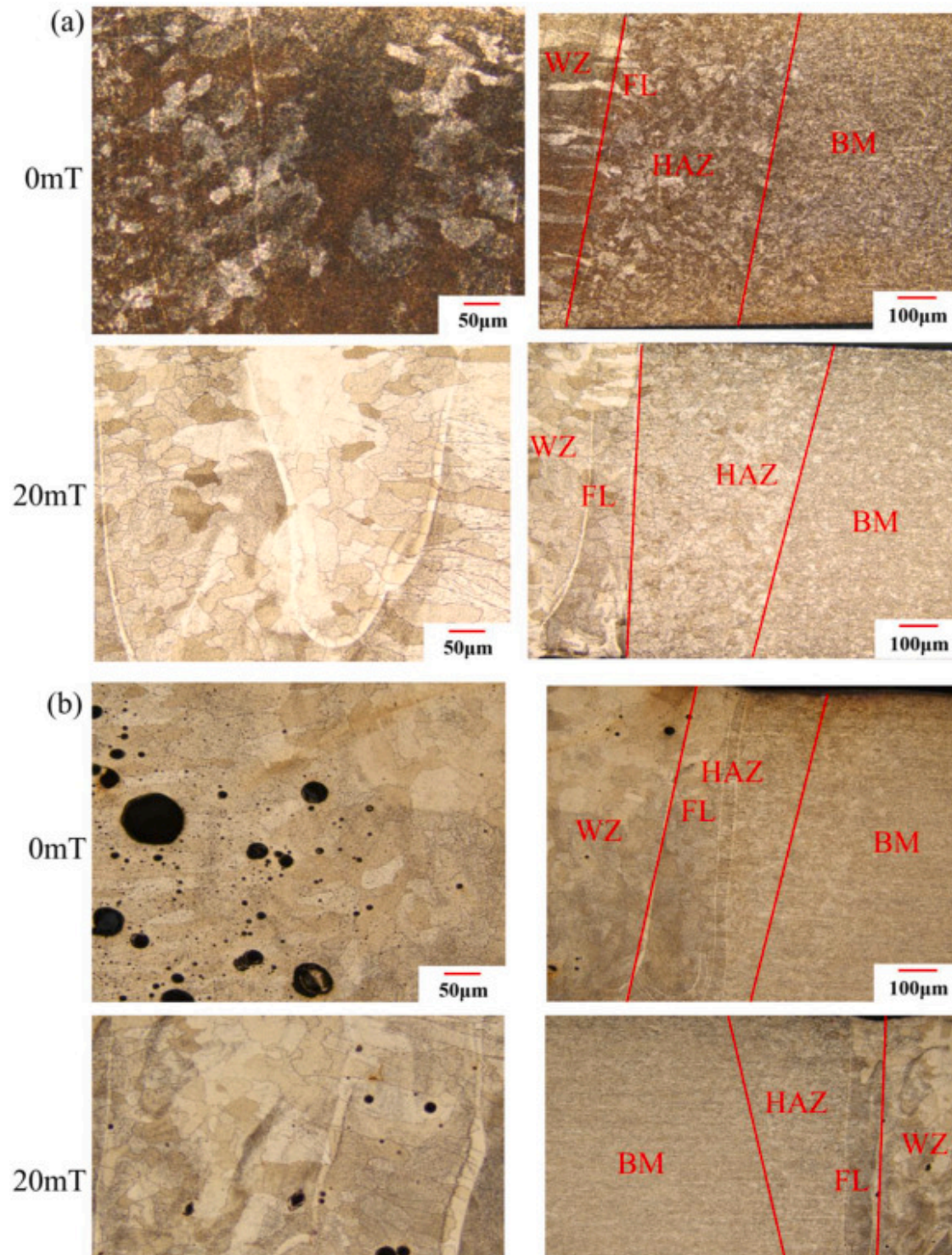
localized perturbations in the flow of the liquid metal, which differs from the smooth flow predicted by the simulation. (2) The physical parameters of the material, such as thermal conductivity, specific heat, and melting point, used in the simulation may deviate from those of the actual material. Additionally, the simulation did not account for the heat dissipation, loss of magnetic field strength due to metal compounds in the molten pool, and the influence of shielding gas on the welding process. (3) The simulation assumes an ideal energy absorption rate for the laser beam, and the model is simplified, not fully accounting for the actual radiation and convection heat dissipation losses. However, in actual welding, the metal vapor and plasma generated on the surface of the molten pool reduce the effective utilization of laser energy, resulting in a smaller melt depth and width in the actual weld compared to the simulation.

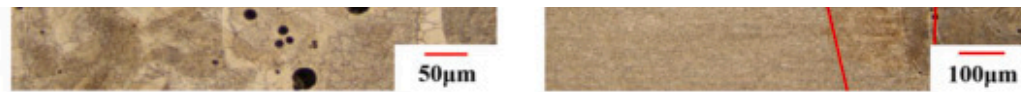
For another, there is a difference between the simulated and experimental weld face wall morphology. The simulation uses a Gaussian-distributed heat source model to model the laser energy input, assuming that the energy is strongest in the center of the spot and gradually decays outward, resulting in a large temperature gradient on both sides of the molten pool, so that the weld face wall shows a rounded transition after cooling and shrinking during the welding process. In the actual welding, due to the laser energy absorption on the material surface of the existence of non-uniformity, and by the plasma scattering, reflection, and other effects, resulting in the energy distribution does not fully comply with the ideal Gaussian distribution, thus making the weld face wall transition is relatively smooth. Additionally, it is assumed in the simulation that the molten pool flow is mainly affected by surface tension, Marangoni convection, and electromagnetic effect, while in the actual welding process, the molten pool flow is controlled by more complicated factors, which not only include the above mechanisms, but also by the complex convection, evaporation driving force, and the combined effect of environmental factors, which further makes the weld face and wall morphology more gentle than the simulation results. Although there are differences between simulation and reality in some parameters and local phenomena, on the whole, the numerical models developed in this study can better predict the fusion morphology of the weld cross-section, and their errors are all controlled within 8 %. This result not only validates the rationality and accuracy of the model but also provides strong theoretical support for optimizing the welding process and selecting parameters.

6.2. Effect of alternating magnetic field on the microstructure of welded joints

Based on the previous analysis, when a magnetic field frequency of 100Hz and a magnetic field strength of 20mT are applied, the weld formation quality is superior compared to other parameter conditions, regardless of whether the weld is penetrated or not. Therefore, the next section will focus on the effect of an unapplied magnetic field versus the application of an alternating

magnetic field with a frequency of 100Hz and a strength of 20mT on the microstructure of both full- or partial-penetration weldments. Fig. 19 shows the microstructure of the joints in both full- or partial-penetration weldments, with and without the application of an alternating magnetic field. From the figure, a welded joint typically consists of three distinct regions: the weld metal zone (WZ), the heat-affected zone (HAZ), and the base metal zone (BM). The columnar crystals in the weld zone dominate; the heat-affected zone near the fusion line shows the α -Cu organization in the annealed state, and the grain size increases compared with the base material, while the base material is typical of the rolled state, and its microstructure consists of fine grains elongated along the rolling direction. In the absence of an applied magnetic field, the weld zone is predominantly composed of horizontally oriented columnar grains on the left and right sides, while the center of the weld contains a relatively small number of equiaxed grains. According to metallurgical principles, the crystal growth mode is primarily influenced by the degree of undercooling of the molten metal. When the undercooling is low, crystals predominantly grow in columnar structures, while when the undercooling is high, equiaxed crystals are more likely to form. Furthermore, combined with the aforementioned analysis of the temperature field of the melt pool under no magnetic field conditions, it can be seen that there is a large temperature gradient in the melt pool, resulting in an uneven temperature distribution, which makes the degree of compositional supercooling relatively smaller, thus contributing to the generation of more columnar dendrites [41]. After the application of magnetic field strength, the number of columnar crystals in the horizontal direction on both the left and right sides of the weld zone of both full- and partial-penetration specimens was significantly reduced, while the number of equiaxed crystals in the center of the weld was significantly increased, accompanied by a refinement in grain size. The reason for this is that after the application of the alternating magnetic field, the molten pool will be subject to the electromagnetic stirring effect generated by the induced Lorentz force, which effectively accelerates the convection of the molten metal, shortens the heat exchange cycle inside the molten pool, makes the temperature distribution more uniform. Consequently, the solid-liquid phase zone expands more rapidly, reducing the temperature gradient at the solidification interface, promotes the nucleation process, improves the stability of the crystallization front, and enables the nuclei inside the molten pool to grow in isotropic direction, which contributes to the formation of equiaxial crystal, and effectively refines the grain structure of equiaxial crystals [42].





[Download: Download high-res image \(2MB\)](#)

[Download: Download full-size image](#)

Fig. 19. Microstructure of full- and partial-penetration joints with or without an alternating magnetic field: (a) full-penetration, (b) partial-penetration.

When the alternating magnetic field is applied, the weld zone microstructure change rule of the joints of the partial-penetration specimen is similar to that of the full-penetration specimen, the columnar crystals on the left and right sides of the weld zone are significantly reduced, and the equiaxed crystals in the center of the weld are significantly increased, and the grain size is refined. However, air holes are still present in the weld of the partial-penetration specimen. Among them, the number and size of porosities are significantly higher when no magnetic field is applied. However, after the application of magnetic field strength, the number of porosities in the weld seam is greatly reduced, with smaller porosities being predominant. This can be attributed to the fact that the electromagnetic stirring effect triggered by the alternating magnetic field can produce stable eddy currents, which can promote Marangoni convection to a certain extent, accelerate the flow of molten metal in the molten pool, help the gas precipitate from the molten metal in the form of bubbles, and reduce the accumulation of gas in the molten pool, thus reducing the porosity of the weld. Additionally, a significant difference in porosity is observed between the weldments in the full- and partial-penetration states. This difference is attributed to the formation of two convection loops within the molten pool during full-penetration welding, which allows gas bubbles to escape from both the top and bottom of the pool simultaneously, which promotes the dynamic opening and closing of the keyhole's bottom, facilitating the ejection of substantial amounts of metal vapor. As a result, the likelihood of keyhole collapse is significantly reduced, enhancing the stability of the molten pool and, consequently, minimizing the formation of porosity [43]. In contrast, during partial-penetration welding, only a single convection loop is formed within the molten pool, with gas bubbles escaping solely from the top. As a result, the metal vapor is primarily ejected through the surface, resulting in a decrease in the stability of the keyhole in the molten pool; some of the gases

in the molten pool are not able to escape in a timely manner, and ultimately these trapped gases solidify within the weld, leading to the formation of porosity.

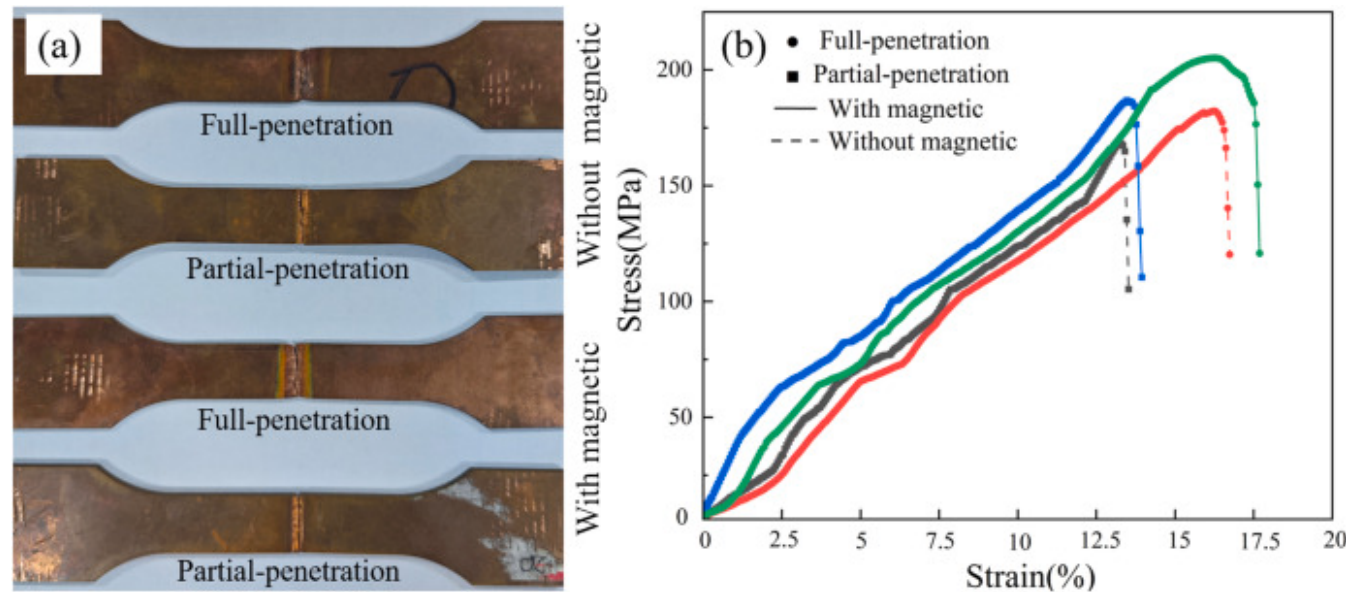
Based on the above analysis, regardless of whether the weld is penetrated or not, the application of the alternating magnetic field significantly reduces the number of columnar crystals in the weld area while notably increasing the number of equiaxed crystals, leading to a refinement of the grain structure. The presence of more small holes in the weld of the partial-penetration specimen, these holes will affect the uniformity of the weld organization, thus reducing the mechanical properties of the weld to a certain extent. In contrast, the absence of porosity defects in the full-penetration specimen and the higher mobility of the molten pool allowed the liquid metal to crystallize more uniformly during the cooling process, thereby reducing segregation and significantly improving the overall performance of the weld. The reason for the more uniform microstructure of the full-penetration specimen is twofold: (1) When full-penetration welding, the molten pool is through the top and bottom, which helps to fully escape the gas and metal vapor, reduce the generation of pores and impurities in the weld, and thus improve the uniformity of the microstructure. (2) The electromagnetic stirring effect caused by the alternating magnetic field in the molten pool forms a spiral eddy current, which enhances the Marangoni convection and makes the temperature diffusion in the molten pool more uniform, contributing to the formation of crystal nuclei at the growth front of columnar crystals, but also hindering the growth of columnar crystals, promoting the formation of equiaxed crystals without obvious orientation and refining the grain structure [44].

6.3. Effect of alternating magnetic field on mechanical properties of welded joints

6.3.1. Tensile strength of welded joints

[Fig. 20](#) shows the fracture morphology and tensile properties of full- and partial-penetration welded joints with or without an alternating magnetic field, and it can be seen from [Fig. 20\(a\)](#) that the specimens all fracture near the fusion zone of the weld. Furthermore, combining [Fig. 20\(b\)](#) with [Fig. 21\(a\)](#), it can be seen that the introduction of an alternating magnetic field significantly increases the tensile strength of the specimen. The joint tensile strength of the full- and partial-penetration welded specimens without magnetic field is 182MPa and 168MPa, respectively, corresponding to only 61.69% and 56.95% of the tensile strength of the base material (295MPa). When the alternating magnetic field was applied, the tensile strength of the full- and partial-penetration weld specimens was 205MPa and 187MPa, respectively; moreover, the tensile strength of the full-penetration weldment increased by 9.63% compared to the partial-penetration weldment. Based on the analysis of weld

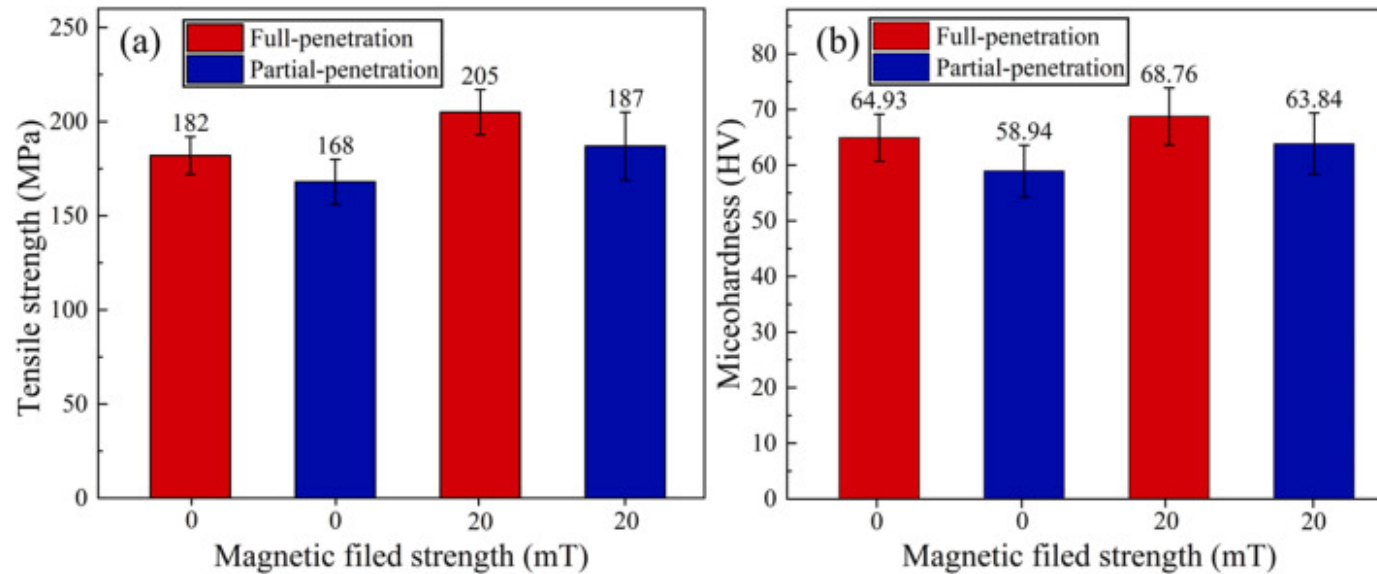
morphology and microstructure of welded parts in the state of full- and partial-penetration, it is that the application of alternating magnetic field effectively improves the stability of the weld pool and the non-uniformity of temperature distribution, reduces the thermal stress generated in the welding process, promotes more uniform solidification and cooling of the weld, and significantly improves the weld forming quality. Simultaneously, the eddy currents enhance the stirring effect within the molten pool, promoting the flow of molten metal, which disrupts and reorganizes the grain structure. This leads to finer and more uniform grains in the weld, thereby improving the tensile strength of the welded joint.



[Download: Download high-res image \(621KB\)](#)

[Download: Download full-size image](#)

Fig. 20. Tensile strength of welded joint specimens with or without an alternating magnetic field: (a) tensile tests samples, (b) stress-strain curves.



[Download: Download high-res image \(375KB\)](#)

[Download: Download full-size image](#)

Fig. 21. Mechanical properties welded joints of full- and partial-penetration specimens with or without an alternating magnetic field: (a) tensile strength, (b) microhardness.

Additionally, under the action of alternating magnetic field of the same frequency and strength, the joint tensile strength of the full-penetration welded part is higher than that of the partial-penetration welded part. The main reason is that the full-penetration welded part can completely melt the substrate, and the laser energy runs through the whole molten pool, which is conducive to the gas generated during the welding process escaping from the molten pool, thereby reducing the formation of pores and improving the tensile strength of the joint. However, in partial-penetration welding, the gas inside the molten pool can only escape from the top of the molten pool, and the gas that fails to escape in time will be solidified inside the weld and form pores, resulting in a decrease in the tensile strength of the joint [45], so there is a significant difference in the tensile strength of the two. The improvement effect of the alternating magnetic field on the full-penetration specimen is significant, and it is better than the partial-penetration specimen in both surface morphology and microstructure. In terms of surface morphology, the weld seam of the full-penetration sample with alternating magnetic field is smoother and the overall consistency is higher, and the forming quality of the full-penetration sample is obviously better than that of the partial-penetration sample. In terms of

microstructure, the grain of the full-penetration sample is significantly refined, and there are no pores and impurities in the weld, which makes the microstructure more uniform, and thus improves the tensile strength of the joint; however, there are defects such as pores in the weld of the partial-penetration sample, resulting in uneven weld structure and reducing the tensile strength of the joint [46].

6.3.2. Microhardness of welded joints

Fig. 21(b) shows the microhardness of the weld zone of the full-penetrated and partial-penetrated specimen joints with or without an alternating magnetic field. As shown in the figure, the microhardness of the weld zones in the full- and partial-penetration specimens without the application of a magnetic field is 64.93 HV and 58.94 HV, respectively, representing only 72.14% and 65.49% of the base metal's microhardness (90 HV). When the alternating magnetic field was applied, the microhardness in the weld zone of the full-penetration and partial-penetration joint specimens was 68.76 HV and 63.84 HV, respectively; moreover, the microhardness of the full-penetration weldments increased by 7.71% compared to that of the partial-penetration weldments. This is attributed to the dominance of columnar crystals in the weld zone of specimens without an applied magnetic field, accompanied by a lower proportion of equiaxed crystals and larger grain sizes, resulting in reduced microhardness in the weld zone. After the application of magnetic field strength, the Lorentz force generated enhances the flow and stirring of the molten metal within the pool. This intensified flow disrupts the local thermal equilibrium of the liquid metal, increases its supercooling, and promotes nucleation and growth in the weld, ultimately resulting in a refined and homogeneous grain structure. The finer grain structure increases dislocation obstruction at grain boundaries, improves the deformation resistance of the weld surface, and thereby enhances the microhardness of the weld zone [47]. Furthermore, under the same alternating magnetic field frequency and strength, the microhardness of the weld zone in the full-penetration specimen is higher than that in the partial-penetration specimen. This is primarily due to the incomplete melting of the base material during the partial-penetration welding process, leading to localized overheating in the weld zone, but the temperature of the surrounding base material changes minimally, resulting in the formation of a pronounced temperature gradient. This uneven temperature distribution leads to an inconsistent cooling rate of the molten pool, which increases the formation of porosity during solidification and ultimately leads to a decrease in the microhardness of the weld. However, in full-penetration welding, the reduced temperature gradient and expanded flow range of the molten pool promote grain refinement in the weld region, leading to a more pronounced increase in weld microhardness.

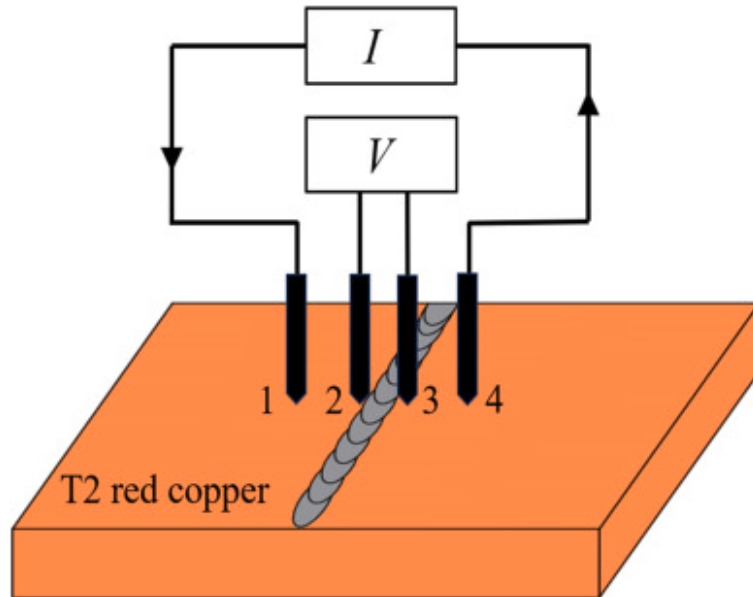
6.4. Effect of alternating magnetic field on the electrical conductivity of welded joints

In order to measure the voltage value of each welded specimen more accurately, the four-point probe method was used to measure the welded specimens (as shown in Fig. 22). The current I flows from pin 1 to pin 4, and the voltage V_{23+} is measured from pins 2 and 3; the current is reversed, flowing from pin 4 to pin 1, and the voltage V_{23-} is measured from pins 2 and 3, and the average of forward and reverse measurements V_{23} is calculated. Similarly, V_{24} is measured according to the $I_{13} V_{24}$ combination, and the geometric correction factor K is determined by the ratio of the two. Finally, according to (15), (16), the sample resistance R and conductivity σ are calculated respectively:

$$R = K \cdot \frac{V}{I} \quad (15)$$

$$\sigma = \frac{L}{RS} \quad (16)$$

Where V is the voltage, I is the current, L is the length of the sample, and S is the cross-sectional area of the material.

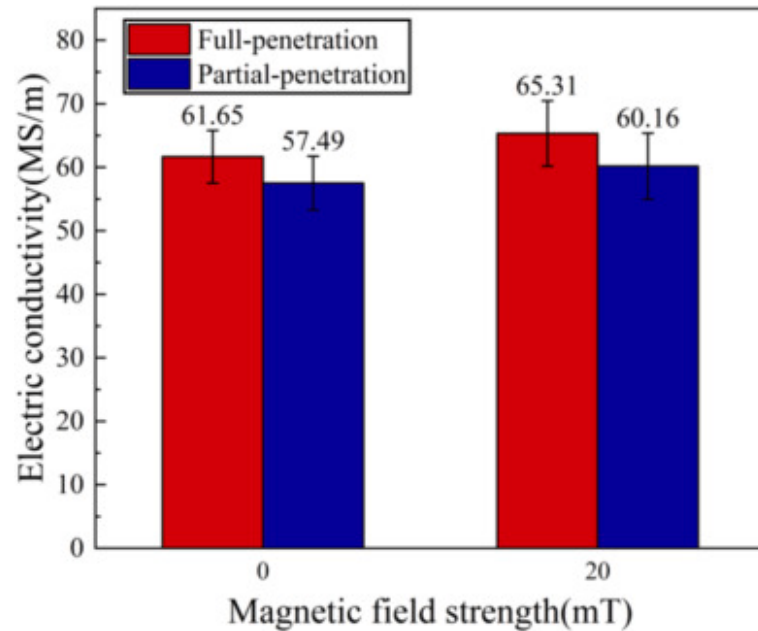


[Download: Download high-res image \(208KB\)](#)

[Download: Download full-size image](#)

Fig. 22. Schematic diagram of four-point probe measurement.

Fig. 23 shows the conductivity of the full- and partial-penetration welds with or without an applied alternating magnetic field. As shown in the figure, the conductivity of full- and partial-penetration welded samples without magnetic field is 61.65 MS/m and 57.49 MS/m, respectively, which is only 78.61% and 73.30% of the conductivity of the base material (78.43 MS/m). When the alternating magnetic field was applied, the conductivity of the full- and partial-penetration weld specimens was 65.31 MS/m and 60.16 MS/m, respectively; moreover, the conductivity of the full-penetration weldment increased by 8.54% compared to the partial-penetration weldment. Combined with the previous analysis of the morphology and microstructure of the weld, the molten pool accelerates the flow of molten metal under the electromagnetic stirring triggered by the Lorentz force, which effectively suppresses the welding defects, enlarges the effective conductive area, and increases the number of free electrons passing through the conductive cross-section per unit of time [48]; secondly, the number of equiaxed crystals in the center of the weld is significantly increased, the number of columnar crystals decreases, and the grain organization is refined considerably. This structure effectively reduces the directional barrier of grain boundaries to the conduction of free electrons, which makes it easier for free electrons to pass through the weld zone, thus reducing the resistivity and improving the conductivity of the weld [49]. Additionally, under the same frequency and strength of alternating magnetic fields, the conductivity of the full-penetration welding specimen is higher than that of the partial-penetration specimen. The reason for this is that the molten pool is in a state of complete penetration during penetration welding, which results in a uniform electromagnetic force acting on the entire molten pool, which helps to enhance the convection of the liquid metal and stabilize the temperature field of the molten pool, resulting in the formation of a uniform and dense grain organization. This fine-grain structure is conducive to electron conduction, reduces resistance scattering, and thus improves the conductivity of the weld. In contrast, in partial-penetration welding, the shallow depth of the molten pool can lead to the formation of an unfused zone or compositional segregation within the weld, resulting in uneven grain growth, an increased number of grain boundaries, and a higher density of dislocations. These factors hinder the flow of current, making it difficult for the conductivity to reach the levels achieved in full-penetration welding.



[Download: Download high-res image \(230KB\)](#)

[Download: Download full-size image](#)

Fig. 23. Conductivity of welded joint of full- and partial-penetration specimens with or without an alternating magnetic field.

7. Conclusion

In this study, the optimal process parameters for alternating magnetic field-assisted laser full- and partial-penetration welding of T2 red copper are first determined. Subsequently, a three-dimensional numerical model is established to investigate its effects on molten pool temperature, fluid flow, and keyhole morphology. Finally, a comparative analysis of the effect of alternating magnetic field on the welding characteristics in terms of microstructure, mechanical properties, and electrical conductivity is presented. The following key conclusions are drawn.

1. Under the influence of a 100Hz, 20mT magnetic field, the quality of weld formation is significantly improved. The weld surface exhibits a uniform and fine fish-scale pattern, while defects such as spatter and humping

are effectively eliminated. Notably, the improvement is more pronounced in full-penetration welds than in partial-penetration welds.

2. After applying the alternating magnetic field, the molten pool exhibited a maximum temperature difference of 871 K under full-penetration conditions, compared to 2982 K in the partial-penetration case. Additionally, the magnetic field enhanced the cross-sectional flow velocity, with peak velocities reaching 2.217 m/s (full-penetration) and 1.932 m/s (partial-penetration). These results demonstrate that the alternating magnetic field has a more pronounced effect on improving temperature uniformity and fluid flow dynamics in the full-penetration state than in the partial-penetration scenario.
3. Compared with partial-penetration welding, the application of an alternating magnetic field more significantly enhances the stability of keyhole depth and morphology in full-penetration welding. This not only ensures a stable fusion depth but also significantly reduces the amplitude of keyhole fluctuations.
4. The cross-section morphology of T2 red copper weld with alternating magnetic field-assisted laser full- and partial-penetration welding is highly consistent with the fusion line morphology of simulation results, with fitting errors of the weld fusion lines all within 8%, thereby verifying the accuracy of the numerical simulation model established in this study.
5. Alternating magnetic field effectively improves the weld morphology and grain structure, welded joints weld zone columnar crystals reduced, more equiaxial crystals and grain refinement, and joints mechanical properties and electrical conductivity have been enhanced. Among them, the microhardness, tensile strength, and conductivity of the weld in the full-penetration state are 68.76 HV, 205 MPa, and 65.31 MS/m, respectively, which are increased by 7.71%, 9.63%, and 8.54% compared to the partial-penetration state. Therefore, the alternating magnetic field has a more pronounced effect on improving the performance of full-penetration welds compared to partial-penetration welds.

CRediT authorship contribution statement

Zhiheng Zeng: Writing – original draft, Software, Investigation. **Lifang Mei:** Writing – review & editing, Validation, Methodology, Data curation. **Dongbing Yan:** Resources, Methodology, Conceptualization. **Qingying Dai:** Writing – review & editing, Validation. **Yang Liu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the Natural Science Foundation of Xiamen (Grant No. [3502Z202373056](#)), Leading Project of Fujian Science and Technology Plan (Grant No. [2022H0055](#)), National Natural Science Foundation of China Project (Grant No. [52375055](#)), Xiamen University of Technology High-level Talent Program (Grant No. [YKJ24036R](#)), and Xiamen University of Technology National Foundation Excellence Program (Grant No. [XPYM2404](#)).

[Recommended articles](#)

Data availability

I have shared the link to my data at the Attach File step

References

- [1] Z.L. Ni, B.H. Li, A.A. Nazarov, *et al.*
Simulation of ultrasonic welding of Cu/Cu joints with an interlayer of Cu nanoparticles
Mater. Today Commun., 39 (2024), Article 109330
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [2] J.Q. Wang, Y.M. Xie, X.C. Meng, *et al.*

Wire-based friction stir additive manufacturing towards isotropic high-strength-ductility Al-Mg alloys

Virtual Phys. Prototyp., 19 (1) (2024), Article e2417369

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [3] J.P. Zheng, J.F. Hu, J.Z. Wang, et al., Progress of laser fusion welding technology of copper and its alloy, Hot Working Technol. 13 (2024) 6-12+29.

[Google Scholar ↗](#)

- [4] Shanshan Tang, Yu Du, Jiale Wang, et al.
Research progress of Al/Cu laser welding for new energy vehicle power battery
Nonferrous Metals Process., 53 (1) (2024), pp. 16-22

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [5] K. Wenchao, Z. Zeng, J.P. Oliveira, et al.
Heat transfer and melt flow of keyhole, transition and conduction modes in laser beam oscillating welding
Int. J. Heat Mass Tran., 203 (2023), Article 123821

[Google Scholar ↗](#)

- [6] Y. Wang, L. Shu, S.N. Geng, et al.
Status and development trend of laser welding technology for automotive body
Chin. J. Lasers, 49 (12) (2022), p. 1202004

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [7] S.Q. Liang, X.D. Gao
Effects of beam oscillation on weld formation and mechanical properties of T2 copper laser lap
J. Netshape Form. Eng., 15 (4) (2023), pp. 131-142

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [8] A. Heider, P. Stritt, A. Hess, et al.
Process stabilization at welding copper by laser power modulation
Phys. Procedia, 12 (2011), pp. 1281-1287

[Google Scholar ↗](#)

- [9] J.K. Jiao, F.Y. Wang, J.Q. Sun, *et al.*
Study on copper surface pre-treating and welding with fiber lasers
Laser Optoelectr. Progress, 53 (3) (2016), Article 031409
[Google Scholar ↗](#)
- [10] B.W. Zhu, Y. Cui, K. Xu, *et al.*
Hard-soft-hard layered structure for enhanced mechanical properties in corrugated rolled AZ31 magnesium alloy sheets
J. Mater. Res. Technol., 35 (2025), pp. 3205-3216
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [11] M.Z. Xin, Z.M. Fan, L.B. Lu, *et al.*
Surface enhancement of metallic alloys by laser precision engineering
Weld. World, 68 (11) (2024), pp. 3033-3050
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [12] Y.L. Zhou, Q.R. Zhang, X. Li, *et al.*
Mechanical performance of laser-textured metallic surface
J. Mater. Res. Technol., 33 (2024), pp. 6084-6089
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [13] H.C. Chen, G.J. Bi, M.L.S. Nai, *et al.*
Influence of surface condition in fiber laser welding of pure copper [C]//International Congress on Applications of Lasers & Electro-Optics
Laser Instit. Am., 2012 (1) (2012), pp. 558-564
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [14] C.Y. Wang, Q. Lin, C. Lu, *et al.*
Blue laser-assisted infrared MOPA laser spot welding of pure copper sheets

Chin. J. Lasers, 49 (16) (2022), Article 1602008

[Google Scholar ↗](#)

- [15] F.Y. Liu, H.Q. Wang, X.Y. Meng, *et al.*
Effect of magnetic field orientation on suppressing porosity in steady-magnetic-field-assisted aluminum alloy deep-penetration laser welding
J. Mater. Process. Technol., 304 (2022), Article 117569
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [16] B.X. Xu, F.Y. Liu, D.Y. Lin, *et al.*
Investigation of weld formation and porosity in 5052 aluminium alloy laser-GMA hybrid welding assisted by magnetic field with different orientations
Sci. Technol. Weld. Join., 27 (8) (2022), pp. 672-682
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [17] L.J. Liang, S.Y. Pang, X.Y. Shao, *et al.*
In situ weak magnetic-assisted thermal stress field reduction effect in laser welding
Metall. Mater. Trans., 49 (1) (2018), pp. 198-209
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [18] L.C. Cao, Q. Zhou, H.P. Liu, *et al.*
Mechanism investigation of the influence of the magnetic field on the molten pool behavior during laser welding of aluminum alloy
Int. J. Heat Mass Tran., 162 (2020), Article 120390
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [19] J.C. Chen, Y.H. Wei, X.H. Zhan, *et al.*
Thermoelectric currents and thermoelectric-magnetic effects in full-penetration laser beam welding of aluminum alloy with magnetic field support
Int. J. Heat Mass Tran., 127 (2018), pp. 332-344
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [20] J.C. Chen, Y.H. Wei, X.H. Zhan, *et al.*
Influence of magnetic field orientation on molten pool dynamics during magnet-assisted laser butt welding of thick aluminum alloy plates
Opt. Laser Technol., 104 (2018), pp. 148-158
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [21] C.J. Hu, F. Yan, Z.W. Zhu, *et al.*
Effects on microstructural refinement of mechanical properties in steel-copper joints laser welded with alternating magnetic field augmentation
Mater. Char., 175 (2021), Article 111059
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [22] F.Y. Liu, C.W. Tan, G.L. Ma, *et al.*
Improvement of inhomogeneity of microstructure and mechanical properties for 316 stainless steel laser-MIG hybrid welded joint assisted by alternating magnetic field
Sci. Technol. Weld. Join., 27 (6) (2022), pp. 437-445
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [23] X.M. Meng, M. Bachmann, A. Artinov, *et al.*
Experimental and numerical assessment of weld pool behavior and final microstructure in wire feed laser beam welding with electromagnetic stirring
J. Manuf. Process., 45 (2019), pp. 408-418
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [24] Z.W. Zhu, X.Q. Ma, C.M. Wang, *et al.*
Modification of droplet morphology and arc oscillation by magnetic field in laser-MIG hybrid welding
Opt Laser. Eng., 131 (2020), Article 106138
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [25] F.Y. Liu, C.W. Tan, L.J. Wu, *et al.*

Influence of waveforms on Laser-MIG hybrid welding characteristics of 5052 aluminum alloy assisted by magnetic field

Opt. Laser Technol., 132 (2020), Article 106508

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [26] M. Bachmann, A. Avilov, A. Gumenyuk, *et al.*
Experimental and numerical investigation of an electromagnetic weld pool control for laser beam welding
Phys. Procedia, 56 (2014), pp. 515-524

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [27] L.F. Mei, Y. Liang, D.B. Yan, *et al.*
Numeric simulation and experimental study in magnetic field-assisted laser scanning welding of T2 red copper

Opt. Laser Technol., 175 (2024), Article 110839

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [28] Z.M. Li, Z.Q. Liu
Comparative study on welding process of red copper with blue diode laser and near infrared fiber laser
J. Netshape Form. Eng., 14 (9) (2022), pp. 142-148

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [29] C.W. Tan, F.Y. Liu, B.X. Xu, *et al.*
Numerical and experimental study of thermal fluid flow and keyhole dynamic in laser welding of aluminum alloy assisted by electromagnetic field

Opt. Laser Technol., 157 (2023), Article 108718

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [30] X. Zhang, Z.Y. Zhao, C.M. Wang, *et al.*
The effect of external longitudinal magnetic field on laser-MIG hybrid welding
Int. J. Adv. Des. Manuf. Technol., 85 (2016), pp. 1735-1743

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [31] A. Evdokimov, N. Doynov, R. Ossenbrink, *et al.*
Thermomechanical laser welding simulation of dissimilar steel-aluminum overlap joints
Int. J. Mech. Sci., 190 (2021), Article 106019
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [32] J.F. Zhou, D w Zhou, J.S. Liu
Numerical and experimental investigation of magnesium/aluminum laser welding with magnetic field
Int. J. Adv. Des. Manuf. Technol., 116 (2021), pp. 545-559
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [33] X.F. Zhou, X.B. Cao, F. Zhang, *et al.*
Numerical and experimental investigation of thermal stress distribution in laser lap welding of Ti6Al4V and 2024 alloy plates
Int. J. Adv. Des. Manuf. Technol., 118 (2022), pp. 1427-1440
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [34] X.S. Gao, C.S. Wu, S.F. Goecke, *et al.*
Numerical simulation of temperature field, fluid flow and weld bead formation in oscillating single mode laser-GMA hybrid welding
J. Mater. Process. Technol., 242 (2016), pp. 147-159
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [35] R.L. Zhang, X.H. Tang, L.D. Xu, *et al.*
Mechanism study of thermal fluid flow and weld root hump suppression in full penetration laser welding of Al alloy with alternating magnetic field support
Int. J. Heat Mass Tran., 166 (2021), Article 120759
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [36] X. Chen, M. Luo, R.Z. Hu, *et al.*
Thermo-electromagnetic effect on weld microstructure in magnetically assisted laser welding of austenite steel

J. Manuf. Process., 41 (2019), pp. 111-118



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [37] T. Tao, D.W. Zhou, J.S. Liu, *et al.*
Improvement of laser welded joint properties of AZ31B magnesium alloy to DP590 dual-phase steel produced by external magnetic field

J. Manuf. Process., 79 (2022), pp. 270-283



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [38] M. Bachmann, V. Avilov, A. Gumenyuk, *et al.*
Experimental and numerical investigation of an electromagnetic weld pool support system for high power laser beam welding of austenitic stainless steel

J. Mater. Process. Technol., 214 (2014), pp. 578-591



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [39] L.F. Mei, Y. Liu, D.B. Yan, *et al.*
Experimental study on magnetic field-assisted laser scanning welding of T2 copper

J. Laser Appl., 36 (2024), Article 012021

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [40] H. Kong, Z.J. Zhao, J.L. Zou, *et al.*
The influence of laser-induced plume in the keyhole on the welding process

Trans. China Weld. Inst., 44 (5) (2023), pp. 20-26+130

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [41] J.S. Cai, X.B. Liu, X.Y. Wang, *et al.*
Simulation of microstructure evolution during solidification process in laser welded molten pool of Al-Cu alloy under forced flow

Mater. Rep., 38 (19) (2024), pp. 228-234

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [42] H.X. Liu, C.X. Cai, Y.H. Jiang, *et al.*
Influence of alternative magnetic field on macro morphology and microstructure of laser cladding Fe-based composite coating
Opt Precis. Eng., 20 (11) (2012), pp. 2402-2410
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [43] L.J. Zhang, J.X. Zhang, A. Gumenyuk, *et al.*
Numerical simulation of full penetration laser welding of thick steel plate with high power high brightness laser
J. Mater. Process. Technol., 214 (8) (2014), pp. 1710-1720
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [44] X.M. Meng, M. Bachmann, A. Artinov, *et al.*
The influence of magnetic field orientation on metal mixing in electromagnetic stirring enhanced wire feed laser beam welding
J. Mater. Process. Technol., 294 (2021), Article 117135
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [45] D.S. Wu, X.M. Hua, L.J. Huang, *et al.*
Elucidation of keyhole induced bubble formation mechanism in fiber laser welding of low carbon steel
Int. J. Heat Mass Tran., 127 (2018), pp. 1077-1086
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [46] C.J. Hu, F. Yan, Y.G. Xu, *et al.*
Effect of alternating magnetic field on microstructure and performance for dissimilar welding-brazing of galvanized steel to 6061 aluminum alloy in an overlapping configuration
Opt. Laser Technol., 155 (2022), Article 108423
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [47] H. Zhu, L. Huang, J.J. Li, *et al.*

Strengthening mechanism in laser-welded 2219 aluminium alloy under the cooperative effects of aging treatment and pulsed electromagnetic loadings

Mater. Sci. Eng., A, 714 (2018), pp. 124-139



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[48] V. Dimatteo, A. Ascari, P. Faverzani, *et al.*

The effect of process parameters on the morphology, mechanical strength and electrical resistance of CW laser-welded pure copper hairpins

J. Manuf. Process., 62 (2021), pp. 450-457



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[49] L. Lu, Y.F. Shen, X.H. Chen, *et al.*

Ultrahigh strength and high electrical conductivity in copper

Science, 304 (2004), pp. 422-426

[View in Scopus ↗](#) [Google Scholar ↗](#)

Cited by (2)

Multiphysics modeling of forced convection in narrow-gap gas metal arc welding: effects of stirring rod geometric configurations

2026, International Journal of Heat and Fluid Flow

[Show abstract](#) ✓

Unraveling keyhole oscillation dynamics in laser processing with multiphysics modeling

2026, International Journal of Mechanical Sciences

[Show abstract](#) ✓

[View Abstract](#)

© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

All content on this site: Copyright © 2026 Elsevier B.V., its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.